

The growth series of the generic right-triangle reflection group is not D -finite

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Abstract

Let $W_{\text{univ}} = \langle R_0, R_1, R_2 \rangle$ be the group generated by the reflections in the sides of a right triangle with generic angle parameter, to which every right triangle group agrees up to a finite, effectively computable depth [3]. Let $U(x) = \sum_n u_n x^n$ be its growth series with respect to the three side reflections (the integer sequence OEIS A396406 [27]), and V its relaxed companion. We prove that U and V continue meromorphically to the disc $|x| < 1$ and have infinitely many poles there, accumulating at the unit circle. Consequently neither is algebraic over $\mathbb{Q}(x)$ or D -finite, so (u_n) and (v_n) are not P-recursive; neither satisfies a linear q -difference equation with Q not a root of unity or a Mahler equation; and, by the Pólya–Bertrandias theorem, $|x| = 1$ is a natural boundary of both. The poles lie over the zeros of the Hahn–Exton q -Bessel function $J_{-1/2}^{(3)}(Z(q); q^2)$, $Z = \sqrt{2(1-q)/q}$, $q = x^2$, which up to a positive factor is the Fredholm determinant of the travel transfer operator; that there are infinitely many of them follows from a spectral argument with no asymptotic estimate. They survive in U and V because the leading Laurent coefficient at each of them is the square of a junction pairing, which has an exact closed form in one bulk quantity and is positive at every travel pole. For the travel resolvent $\Sigma_0/(1 - \Sigma_1)$ all of this is unconditional. For U and V it rests on the transfer model, a theorem whose assembly step is a hand proof, and on the junction-pairing theorem, whose positivity uses an analytic gate estimate proved partly by verified interval arithmetic. The long proofs are given in a companion paper.

1 Introduction

Growth series of groups. For a group G with a finite generating set S , the growth series $\sum_{n \geq 0} a_n x^n$ counts the elements of G by their word length in S . In the classical families it is rational: hyperbolic groups have rational growth series with respect to every finite generating set (Cannon [8]), and Coxeter systems are governed by Steinberg’s rational formula [15, §5.12]. For wreath products, including lamplighter-type groups, Parry computed the growth series for natural generating sets and found them *algebraic* [24]. Provably transcendental growth series are rarer. Stoll [30] showed that the higher Heisenberg group H_5 has a transcendental growth series with respect to its standard generating set, and a rational one with respect to a specially chosen generating set; the three-dimensional Heisenberg group, by contrast, has rational growth for every finite generating set (Duchin–Shapiro [11]). Two general mechanisms give growth series that are not even D -finite. For groups of polynomial growth, and for groups of intermediate growth such as Grigorchuk’s [13], the growth series has radius of convergence 1 and integer coefficients, so by Pólya–Carlson [10] it

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is rational or has $|x| = 1$ as a natural boundary; in the second case, which includes Stoll's series for H_5 and every group of intermediate growth, it is not D -finite. And a finitely presented group with unsolvable word problem has a non-computable growth function, while a D -finite series with rational coefficients has computable coefficients, so its growth series is not D -finite either.

The group. A right triangle T has three sides; the reflections across them generate a group $W_T = \langle R_0, R_1, R_2 \rangle \leq \text{Aff}(\mathbb{R}^2)$. For a right triangle whose angle parameter is generic, W_T is the *universal* group W_{univ} of [3], and every other right triangle group agrees with W_{univ} in the ball of some finite radius and deviates beyond it; the radius is effective [3, Theorem “Effective universality”, thm:effective-universality] (for the $(1, 2)$ triangle the first deviation is at depth 33, [3, Theorem “Sharp uniform universality at depth 32”, thm:universality-sharp]). Irrationality of the angles alone (Niven [21]) does not give genericity: a triangle in $\mathbb{Q}^2$ with unequal legs has irrational angles, yet it is not universal. Throughout, $U(x) = \sum_n u_n x^n$ is the growth series of W_{univ} with respect to $\{R_0, R_1, R_2\}$. The sequence $u_n = 1, 3, 5, 8, 13, 21, 34, 55, 89, \dots$ is OEIS A396406; it coincides with the Fibonacci numbers F_{n+3} for $1 \leq n \leq 9$ and deviates at $n = 10$. W_{univ} is not a Coxeter group [16], and it is virtually the lamplighter $\mathbb{Z} \wr \mathbb{Z}$ [3]. Its generating set is not constructed but imposed by Euclidean geometry. For a group generated by reflections, Steinberg's formula would make the growth series rational in the Coxeter case, and Parry's theorem makes the natural wreath-product series algebraic; the result below places this geometric system outside both classes.

The word length $\ell_T(g)$ is realised by strand walks in a finite transition system; permitting free isolated cycles yields a smaller *relaxed* length $\ell_R(g)$, with $\ell_T = \ell_R + 2c(g)$, $c(g) \in \mathbb{Z}_{\geq 0}$ a cycle count. The relaxed counting function is the companion V ; one has $v_n \geq u_n$, with equality for $n \leq 7$ and strict inequality at $n = 8$.

The main theorem.

Main Theorem. (i) U and V continue to single-valued meromorphic functions on the disc $|x| < 1$, and each has infinitely many poles there, accumulating at the unit circle (Lemma 13, Theorems 18 and 20).

(ii) Consequently U and V are transcendental over $\mathbb{Q}(x)$ and not D -finite, so (u_n) and (v_n) are not P -recursive (Theorem 26); neither satisfies a linear q -difference equation $\sum_i a_i(x)f(Q^i x) = b(x)$ with Q not a root of unity, nor a Mahler equation (Proposition 28); and $|x| = 1$ is a natural boundary of both (Theorem 31).

(iii) The travel resolvent $G_0^T = \Sigma_0/(1 - \Sigma_1) \in \mathbb{Z}[[q]]$ has all of these properties in the variable q , with natural boundary $|q| = 1$ (Theorems 17, 26, 31, Proposition 28).

(iv) The exponential growth rate of (u_n) and of (v_n) is $\beta_2 = 1.4916177871\dots$, where β_2^{-2} is the smallest zero in $(0, 1)$ of $q \mapsto J_{-1/2}^{(3)}(Z(q); q^2)$ (Corollary 19).

(Standing: (iii) is unconditional. (i), (ii) and (iv) rest on Theorems 12 and 15, as stated there.)

The mechanism. The proof does not go through the coefficients. It locates infinitely many poles of U and V inside the disc $|x| < 1$ of meromorphy, beyond the radius of convergence $1/\beta_2 = 0.6704\dots$, and then applies three classical facts: an algebraic function has finitely many poles in the disc; a pole of a solution of a linear differential equation whose coefficients are holomorphic near the closed disc is a zero of its leading coefficient (Lemma 25); and an integer power series that is meromorphic on a simply connected domain of conformal radius larger than 1 is rational (Pólya–Bertrandias, Theorem 30). The poles are produced in three steps.

(1) *The transfer model.* Grading by relaxed length and marking cycles gives a bivariate series $\mathcal{W}(x, y)$ with $V = \mathcal{W}(x, 1)$ and $U = \mathcal{W}(x, x^2)$. Theorem 12 writes $\mathcal{W}$ as a sum of three sectors. Two of them are pairings against the resolvent $(I - \mathcal{T})^{-1}$ of the *travel operator* $\mathcal{T} = D_e K$, with $q = x^2$, $K = [q^{\max(s, s')}]_{s, s' \geq 0}$ and $D_e = \text{diag}(2q^{1+s})$; the bulk enters only through explicit vectors built from the resolvent of the same semiseparable kernel on a shifted index set, with a rank-one gap term, and the third sector contains no travel resolvent. Lemma 13 continues $\mathcal{W}$ meromorphically to $|x| < 1$.

(2) *The travel poles.* The poles of $(I - \mathcal{T})^{-1}$ are the zeros of its Fredholm determinant, and Theorem 3 computes that determinant in closed form:

$$\det(I - \lambda \mathcal{T}) = \sum_{k \geq 0} \frac{(-2\lambda(1 - q))^k q^{k^2}}{(q; q)_{2k}} = \cos(\sqrt{\lambda} Z; q^2) = c_q(\sqrt{\lambda} Z)^{1/2} J_{-1/2}^{(3)}(\sqrt{\lambda} Z; q^2),$$

the q -cosine of Koornwinder and Swarttouw [18] and the Hahn–Exton q -Bessel function [14, 12, 17]. So the travel poles are the $q \in (0, 1)$ with $J_{-1/2}^{(3)}(Z(q); q^2) = 0$ (Corollary 5). A symmetrisation of $\mathcal{T}$ is a positive trace-class operator with simple eigenvalues $1/\lambda_k(q)$; the branches satisfy $\lambda_1(q) > (1 - q^2)/(2q)$ and $\lambda_k(q)/(1 - q) \rightarrow (2k - 1)^2 \pi^2/8$, so each crosses the level 1, and distinct branches cross at distinct points: there are infinitely many travel poles (Proposition 6), locally finite in $(0, 1)$ (Lemma 7), hence accumulating at $q = 1$. This step uses no asymptotic estimate of a q -series.

(3) *The residue.* At a travel pole q_m the leading Laurent coefficient of $\mathcal{W}(\cdot, y)$ is $c_\star \Pi_y(q_m)^2/(2x_m)$, where $\Pi_y = \langle \lambda^{(y)}, R \rangle$ pairs the junction vector with the null vector R of $I - \mathcal{T}(q_m)$ (Proposition 14). The junction is not of rank one, so the numerator does not factor through the total bulk sum; nevertheless two shift identities for R collapse the pairing to an exact closed form in the single bulk quantity t_1 (Theorem 15), for instance

$$(1 - \mathfrak{g}_V t_1) \Pi_1(q_m) = \frac{2q(1 + x)}{1 - q} \left(t_1 + \frac{1 + x}{2x} \right),$$

and this is positive at every travel pole: analytically for $\tau_m \leq 5 \cdot 10^{-3}$ once the *gate* $|\mathfrak{g}_V t_1| < 1$ holds, and at the first twelve poles by an interval certificate. For the travel resolvent the corresponding non-vanishing is an exact identity, $\Sigma_0(q_m) S_0(q_m) = 2q_m/(1 - q_m)$ (Proposition 11), a consequence of the q -Pythagorean identity at a zero of the q -cosine.

Standing, and what is cited from the companion paper. The statements about the travel resolvent are unconditional. The statements about U and V rest on two theorems. The transfer model (M), Theorem 12: its metric identity and the faithfulness of the model are formalised in Lean 4, and its block decomposition (M3') is a hand proof, given in the companion paper [5], checked against exact enumeration to x^{26} and not formalised. The junction pairing (R-J), Theorem 15: its closed form is proved here; its positivity is analytic for $\tau_m \leq 5 \cdot 10^{-3}$, given the gate, and rests on an interval-arithmetic certificate at the first twelve poles; only the analytic range, which contains all but finitely many travel poles, is needed for (i)–(ii) of the Main Theorem. The gate itself, the amplitude estimate ($\star$) and the lower bound on the bulk denominator at the travel poles, is proved in [5], with constants produced by committed certificates and re-certified in directed-rounding interval arithmetic on the range stated there.

This paper states and proves the spectral and q -trigonometric steps, the local form, the closed form of the junction pairing and every consequence above, and it states the transfer model with an outline of its proof. The long technical proofs are in [5] and are cited by theorem label: the block assembly (the section identity, the telescoping of the travel and bulk sums, the reciprocity and the Möbius factorisation of the gap-marked bulk resolvent), the site cost of the crossing optimisation and

its Lean formalisation, the gap-free theorem behind Proposition 1, the proof of (M3'), the gate with its certificates, the explicit lower bound for S_e at the travel poles, an absolute bound on an auxiliary contour integral T_2 which gives a second proof of the infinitude of the travel poles and localises them in windows of w (the window statement is used only in the Mahler part of Proposition 28), the Pólya–Carlson dichotomy for the four building blocks, the spectral sign laws with their standing, and the value-transcendence question for the number β_2 . The archival full version [4] contains both papers; labels are shared with it.

Related results. The cogrowth series is settled by general theory: W_{univ} is amenable, being virtually metabelian, and not virtually nilpotent, so its cogrowth series is not D -finite by Bell–Mishna [1]. In higher dimensions the picture is subtler [6]: the abstract right-angled Coxeter envelope of an n -orthoscheme has rational growth, but the geometric group is an amenable proper quotient that matches this envelope only to a finite depth. Whether U and V are D -algebraic is open.

Organisation. Section 2 introduces the building blocks and the travel poles. Section 3 identifies the pole equation with the q -cosine and with the Fredholm determinant of the travel operator, proves that the travel poles are infinite and accumulate only at $q = 1$, and proves that the numerators do not vanish there. Section 4 states the transfer model, continues $\mathcal{W}$ meromorphically and computes the shape of its local form at a travel pole; Section 4.3 evaluates the junction pairing. Section 5 proves transcendence, and Section 6 the statements beyond algebraicity. Section 7 lists open questions.

2 The catalytic equation and its building blocks

Write $q = x^2 = e^{-\tau}$, $w = \sqrt{2/\tau}$. The relaxed length statistic satisfies a linear q -difference equation with dilation $t \mapsto q^2 t$; telescoping it produces four integer power series in q ,

$$\Sigma_1(q) = \sum_{i \geq 1} (-1)^{i-1} \frac{w^{2i}}{(2i)!} \eta_i, \quad S_e(q) = \sum_{j \geq 0} \frac{(-2(1-q))^j q^{j(j+1)}}{(q; q)_{2j}}, \quad (1)$$

and their companions Σ_0, S_0 , where $\eta_j = e^{-B_j}$ is the Gaussian-decaying form factor of [5, §sec:sd] (equivalently $\eta_j = q^{j^2+j} [2(1-q)]^j (2j)! / [(q; q)_{2j} W^{2j}]$ with $W = w e^{-\tau/2}$; the identification of this display with $1 - \Sigma_1 = \cos(Z; q^2)$ of Proposition 2 is the collapse used in [5, Lemma “infinitude and accumulation of the travel poles”, lem:infpoles]). We write $\Sigma_1^T = \Sigma_1$ when the travel rôle is to be emphasised. The relaxed travel resolvent is $G_0^T = \Sigma_0 / (1 - \Sigma_1)$ ([5, Proposition “the travel k -sum telescopes”, prop:travelsum]), and U inherits its denominator $1 - \Sigma_1$ (Proposition 1); how the two blocks assemble into V and U , and how far that assembly is proved, is [5, §sec:assembly]. The *travel poles*

$$q_1 < q_2 < \cdots \uparrow 1, \quad \Sigma_1^T(q_m) = 1,$$

form an infinite set accumulating at $q = 1$ (Proposition 6 with Lemma 7; a second proof, through an absolute bound on an auxiliary contour integral, is [5, Lemma “infinitude and accumulation of the travel poles”, lem:infpoles]).

The four blocks are the k -recursion sums

$$\Sigma_k = \sum_{j \geq 0} A_{k+2j} \prod_{i < j} C_{k+2i}, \quad S_k = \sum_{j \geq 0} \alpha_{k+2j} \prod_{i < j} \gamma_{k+2i}, \quad (2)$$

$$A_k = \frac{2q}{1 - q^{k+1}}, \quad C_k = \frac{2q^{k+3}}{1 - q^{k+2}} - \frac{2q^{k+2}}{1 - q^{k+1}}, \quad \alpha_k = \frac{2q^{k+1}}{1 - q^{k+1}}, \quad \gamma_k = \alpha_{k+1} - \alpha_k,$$

Σ_0, Σ_1 carrying the travel rôle and S_0, S_1 the bulk rôle; Σ_1 and S_1 are the denominators of (1) and Σ_0, S_0 their numerators. Every factor is a Lambert series with integer coefficients, and the j -th summand of Σ_1 starts at order $(j+1)^2$, so each coefficient is a finite integer sum; $\Sigma_1 = 2q + 2q^3 - 4q^4 + 6q^5 - \dots$. All four blocks are holomorphic on $|q| < 1$: by Proposition 2 and Lemma 9 below each is a q -trigonometric series whose j -th term carries the Gaussian factor $q^{j^2+O(j)}$ against a denominator $(q; q)_{2j}$ or $(q; q)_{2j+1}$, bounded below by $|(q; q)_\infty| > 0$ on compact subsets of the disc (this is the first clause of [5, Theorem thm:blocks], where the Pólya–Carlson dichotomy for the four blocks is proved). The coefficients of G_0^T are the travel-run counts 2, 6, 10, 26, 54, 114, 274, $\dots$, integers.

Proposition 1 (Invariance of the travel block). *The denominator $1 - \Sigma_1$ is the identical, cycle-marker-free object in the generating functions of U and of V ; hence U and V share the travel poles q_m .*

Proof. A pure-travel element has no gap edge, since its span is I_k and $f_j = \pm 1$ there (I_k the travel interval and f_j the travel indicator of [5, §sec:sitecost]; this is [5, Corollary cor:localzero]), so [5, Theorem “no gap edge forces $c = 0$ ”, thm:nogap] gives $c = 0$: some relaxed-optimal realisation has no isolated cycle. That realisation is a single open walk, hence admissible for ℓ_T , so $\ell_T \leq \ell_R$; and $\ell_T \geq \ell_R$ because the relaxed minimum ranges over a superset of the realisations. Therefore $\ell_T = \ell_R$ on pure-travel elements and the two gradings of T agree termwise. The travel recursion is assembled from pure-travel runs alone, appending one $f = \pm 1$ edge at a time, so it is computed in the sector $c \equiv 0$ and the travel block carries y^0 only. $\square$

The history of this proposition, which earlier versions carried as the hypothesis (T), is recorded in [5, §sec:inputT].

3 The travel poles

3.1 The travel-pole equation as a q -trigonometric identity

The two series that govern the poles admit an exact identification with the q -cosine of Koornwinder and Swarttouw [18], $\cos(z; q^2) := {}_1\phi_1(0; q; q^2, q^2 z^2) = \sum_{j \geq 0} (-1)^j q^{j^2+j} z^{2j} / (q; q)_{2j}$.

Proposition 2. *Let $z_0 = \sqrt{2(1-q)}$ and $Z = z_0/\sqrt{q}$. Then*

$$1 - S_1 = S_e = \cos(z_0; q^2), \quad 1 - \Sigma_1 = \cos(Z; q^2).$$

In particular the travel poles q_m are exactly the solutions of $\cos(Z; q^2) = 0$, and S_e is the same function evaluated one half q -step below its own zero.

Proof. Collapse the k -recursion (2) on its two odd ladders, the travel one carrying Σ_1 and the bulk one carrying S_1 . Write $1 - q^m = 2e^{-m\tau/2} \sinh(m\tau/2)$ in the sinh-forms of the two series ([5, §sec:sd]): the products collapse to $1 - \Sigma_1 = \sum_j (-1)^j (2(1-q))^j q^{j^2} / (q; q)_{2j}$ and $1 - S_1 = \sum_j (-1)^j (2(1-q))^j q^{j^2+j} / (q; q)_{2j}$, the second being S_e as defined in (1). Comparing with the series for $\cos(z; q^2)$ term by term forces $q^2 z^2 = 2(1-q)q$, i.e. $z = Z$, in the first case and $q^2 z^2 = 2(1-q)q^2$, i.e. $z = z_0$, in the second; both series converge for $|q| < 1$, the factor q^{j^2} dominating. Both collapses are recomputed as identities of integer power series, to degree 1160 in `blocks_growth.py` and at two truncation degrees in `f2_pole_route.py`. The per-term identity is machine-checked in Lean 4 (`GramCasoratian.lean`, theorem `sigma1_term`; axioms `propext`, `Classical.choice`, `Quot.sound`, no sorry). $\square$

The Hahn–Exton normalisation. Write $J_\nu^{(3)}(x; q^2) = x^\nu \frac{(q^{2\nu+2}; q^2)_\infty}{(q^2; q^2)_\infty} {}_1\phi_1(0; q^{2\nu+2}; q^2, q^2 x^2)$ for the Hahn–Exton q -Bessel function, in the normalisation of [17, eq. (2.4)] with base q^2 , and put

$$c_q := \frac{(q^2; q^2)_\infty}{(q; q^2)_\infty} > 0.$$

Since $(q^2; q^2)_k (q; q^2)_k = (q; q)_{2k}$ and $(q^2; q^2)_k (q^3; q^2)_k = (q; q)_{2k+1} / (1 - q)$, comparing coefficients gives, for $x > 0$,

$$\cos(x; q^2) = c_q x^{1/2} J_{-1/2}^{(3)}(x; q^2), \quad \sin(x; q^2) = c_q x^{1/2} J_{1/2}^{(3)}(x; q^2).$$

So the travel poles are the zeros in $(0, 1)$ of $q \mapsto J_{-1/2}^{(3)}(Z(q); q^2)$ (Corollary 5); [5, Proposition “structure of the zero curve”, prop:zerostructure(i)] records the same identity in the variable of [5, Appendix app:beta2].

3.2 The pole equation as a Fredholm determinant

The q -cosine of Proposition 2 is the Fredholm determinant of the travel operator. Throughout this subsection $0 < q < 1$, $D_e = \text{diag}(2q^{1+s})_{s \geq 0}$ and $K = [q^{\max(s, s')}]_{s, s' \geq 0}$, so that $\mathcal{T} = D_e K$ is the travel operator of [5, Definition “section identity”, def:model]. Put

$$\mathbf{S} := D_e^{1/2} K D_e^{1/2}, \quad \mathbf{S}[s, s'] = 2q^{1+(s+s')/2+\max(s, s')}, \quad F(\lambda, q) := \sum_{k \geq 0} \frac{(-2\lambda(1-q))^k q^{k^2}}{(q; q)_{2k}} \quad (\lambda \in \mathbb{C}).$$

The spectral parameter λ is not to be confused with the junction vectors $\lambda^{(y)}$ of §4.

Theorem 3 (the travel determinant). *Let $0 < q < 1$ and $Z = \sqrt{2(1-q)}/q$.*

- (i) $\mathbf{S}$ is a positive definite trace-class operator on $\ell^2(\mathbb{Z}_{\geq 0})$, with $\text{tr } \mathbf{S} = 2q/(1 - q^2)$.
- (ii) $\mathcal{T}$ and $\mathbf{S}$ have the same principal minors, $\det \mathcal{T}[I, I] = \det \mathbf{S}[I, I]$ for every finite $I \subset \mathbb{Z}_{\geq 0}$, and the same non-zero eigenvalues, the eigenvectors corresponding by $R = D_e^{1/2} \phi$: if $\mathbf{S} \phi = \kappa \phi$ with $\phi \in \ell^2$ and $\kappa \neq 0$, then $R \in \ell^1$ and $\mathcal{T} R = \kappa R$; conversely, if $R \in \ell^1$ and $\mathcal{T} R = \kappa R$ with $\kappa \neq 0$, then $\phi = D_e^{-1/2} R \in \ell^2$ and $\mathbf{S} \phi = \kappa \phi$.
- (iii) For every $\lambda \in \mathbb{C}$, with absolutely convergent series,

$$\det(I - \lambda \mathbf{S}) = \sum_{k \geq 0} (-\lambda)^k \sum_{|I|=k} \det \mathbf{S}[I, I] = F(\lambda, q),$$

$$F(\lambda, q) = \cos(\sqrt{\lambda} Z; q^2) = c_q (\sqrt{\lambda} Z)^{1/2} J_{-1/2}^{(3)}(\sqrt{\lambda} Z; q^2),$$

the last two expressions being even in $\sqrt{\lambda}$. We write $\det(I - \lambda \mathcal{T})$ for the same Fredholm series formed with the minors of $\mathcal{T}$; by (ii) it is the same function.

- (iv) The eigenvalues of $\mathbf{S}$ are $1/\lambda_k(q)$, $k \geq 1$, with $0 < \lambda_1(q) < \lambda_2(q) < \dots \rightarrow \infty$, all simple; $\sum_k 1/\lambda_k(q) = 2q/(1 - q^2)$ and $F(\lambda, q) = \prod_{k \geq 1} (1 - \lambda/\lambda_k(q))$.
- (v) $1 - \Sigma_1 = F(1, q) = \det(I - \mathcal{T})$, and $S_e = F(q, q) = \det(I - q\mathcal{T}) = \det(I - M_0)$: the bulk operator M_0 of [5, Definition “section identity”, def:model] is $q\mathcal{T}$ after the index shift $b = s + 1$.

Proof. (i) For $x \in \ell^2$ put $X = D_e^{1/2}x$, which lies in ℓ^1 by Cauchy–Schwarz. Since $q^{\max(s,s')} = (1-q) \sum_{t \geq \max(s,s')} q^t$,

$$\langle x, \mathbf{S}x \rangle = \langle X, KX \rangle = (1-q) \sum_{t \geq 0} q^t \left| \sum_{s \leq t} X_s \right|^2 \geq 0,$$

the double sums converging absolutely, with equality only if every partial sum of X vanishes, that is $X = 0$, that is $x = 0$. $\mathbf{S}$ is Hilbert–Schmidt, hence bounded, and it is positive definite; a positive operator with summable diagonal is trace class, with trace $\sum_s 2q^{1+s}q^s = 2q/(1-q^2)$.

(ii) $\mathcal{T}[I, I] = D_e[I, I]K[I, I]$ and $\mathbf{S}[I, I] = D_e[I, I]^{1/2}K[I, I]D_e[I, I]^{1/2}$ with $D_e[I, I]$ diagonal, so the two determinants agree. If $\mathbf{S}\phi = \kappa\phi$, then $R = D_e^{1/2}\phi \in \ell^1$ and $\mathcal{T}R = D_e^{1/2}\mathbf{S}\phi = \kappa R$. Conversely, $\mathcal{T}R = \kappa R$ gives $R = \kappa^{-1}D_e(KR)$ with $|(KR)_s| \leq \|R\|_{\ell^1}$, so $\phi = D_e^{-1/2}R = \kappa^{-1}D_e^{1/2}KR \in \ell^2$ and $\mathbf{S}\phi = D_e^{1/2}KR = \kappa\phi$. The operators are not boundedly similar, $D_e^{-1/2}$ being unbounded; only (ii) is used.

(iii) For trace-class A , $\det(I - \lambda A) = \sum_k (-\lambda)^k \text{tr } \Lambda^k(A)$ [26, Ch. 3] (the von Koch form; compare Fredholm’s formula for continuous kernels, [26, Thm. 3.10]), and computing the trace of $\Lambda^k(\mathbf{S})$ in the orthonormal basis $e_{s_1} \wedge \cdots \wedge e_{s_k}$, $s_1 < \cdots < s_k$, whose diagonal entries are the principal minors, gives the first equality. The principal minors of the positive $\mathbf{S}$ are ≥ 0 , so the inner sums converge absolutely, to the values computed next, and the outer series then converges absolutely for every λ . For $I = \{s_1 < \cdots < s_k\}$ the matrix $K[I, I]$ is $(a_{\max(i,j)})$ with $a_i = q^{s_i}$ decreasing; subtracting row $i+1$ from row i for $i < k$ leaves a lower triangular matrix with diagonal $a_1 - a_2, \dots, a_{k-1} - a_k, a_k$, so $\det K[I, I] = q^{s_k} \prod_{i < k} (q^{s_i} - q^{s_{i+1}})$ and, with the gaps $g_i = s_{i+1} - s_i \geq 1$,

$$\det \mathbf{S}[I, I] = 2^k q^k q^{2(s_1 + \cdots + s_k)} \prod_{i < k} (1 - q^{g_i}).$$

Since $s_1 + \cdots + s_k = ks_1 + \sum_{j < k} (k-j)g_j$, the sum over $s_1 \geq 0$ and $g_1, \dots, g_{k-1} \geq 1$ factorises; with $\sum_{g \geq 1} q^{2rg}(1-q^g) = q^{2r}(1-q)/((1-q^{2r})(1-q^{2r+1}))$,

$$\sum_{|I|=k} \det \mathbf{S}[I, I] = \frac{(2q)^k}{1-q^{2k}} \prod_{r=1}^{k-1} \frac{q^{2r}(1-q)}{(1-q^{2r})(1-q^{2r+1})} = \frac{(2(1-q))^k q^{k^2}}{(q; q)_{2k}}.$$

Hence $\det(I - \lambda \mathbf{S}) = F(\lambda, q)$; the case $k = 1$ is the trace. Finally $(2\lambda(1-q))^k q^{k^2} = q^{k^2+k}(\sqrt{\lambda}Z)^{2k}$, since $Z^2 = 2(1-q)/q$, so $F(\lambda, q) = \cos(\sqrt{\lambda}Z; q^2)$, and the Hahn–Exton form is the normalisation after Proposition 2.

(iv) For positive trace-class $\mathbf{S}$ the determinant is the product $\prod_k (1 - \lambda \kappa_k)$ over its eigenvalues $\kappa_k > 0$ counted with multiplicity [26, Ch. 3], so the $1/\kappa_k$ are the zeros of $F(\cdot, q)$, the multiplicity of κ_k being the order of the zero. By (iii) these are the $\lambda = x^2/Z^2$ with $x > 0$ a zero of $\cos(x; q^2) = c_q x^{1/2} J_{-1/2}^{(3)}(x; q^2)$ ($\cos(\cdot; q^2)$ is even with $\cos(0; q^2) = 1$). For $\nu > -1$ the zeros of $J_\nu^{(3)}(\cdot; q^2)$ are real and simple and form an infinite sequence [17, Cor. 3.2, Lemma 3.3, Thm. 3.4]; here $\nu = -\frac{1}{2}$ with base q^2 . So the eigenvalues are simple and infinitely many, and the sum of the $1/\lambda_k$ is the trace.

(v) At $\lambda = 1$, $F(1, q) = \sum_k (-1)^k (2(1-q))^k q^{k^2} / (q; q)_{2k} = 1 - \Sigma_1$ (Proposition 2); at $\lambda = q$, $F(q, q) = \sum_k (-2(1-q))^k q^{k^2+k} / (q; q)_{2k} = S_e$ by (1). And $M_0[s+1, s'+1] = 2q^{s+1}q^{\max(s,s')+1} = q\mathcal{T}[s, s']$, so M_0 has the principal minors of $q\mathcal{T}$. $\square$

The identity $\det(I - \lambda \mathbf{S}) = F(\lambda, q)$ was checked in double precision against truncated matrices at $q = 0.3, 0.6, 0.8$ and four values of λ (one of them non-real), to within $3.2 \cdot 10^{-15}$, and the gap sum against a brute-force sum of minors for $k \leq 3$ at $q = 0.3$.

Remark 4 (a q -string). With the Jackson integral $\int_0^1 f d_q t = (1-q) \sum_{s \geq 0} q^s f(q^s)$, let $K_q f(x) = \int_0^1 \min(x, t) f(t) d_q t$ on the lattice $x \in \{q^s\}$. At $x = q^s$, $(KD_e u)_s = \sum_{s'} q^{\max(s, s')} 2q^{1+s'} u_{s'} = \frac{2q}{1-q} (K_q u)(q^s)$, that is,

$$\mathcal{T}^\top = KD_e = \frac{2q}{1-q} K_q,$$

so up to the transposition, which changes neither the principal minors nor the eigenvalues, $\mathcal{T} = \frac{2q}{1-q} K_q$. The kernel $\min(x, t)$ is the Green function of $-d^2/dx^2$ on $[0, 1]$ with a Dirichlet end at 0 and a Neumann end at 1: $\mathcal{T}$ is the compliance operator of a q -string with uniform Jackson mass, fixed at 0 and free at 1, and $u = KR$ is its displacement ($u_s \rightarrow 0$ as $x = q^s \rightarrow 0$). As $q \rightarrow 1$ the kernel tends to the classical one, with eigenvalues $4/((2k-1)^2 \pi^2)$; this is the heuristic behind $\lambda_k(q) \sim (1-q)(2k-1)^2 \pi^2/8$, proved for fixed k in Proposition 6.

Corollary 5 (the travel poles are Hahn–Exton zeros). *A point $q \in (0, 1)$ is a travel pole if and only if*

$$J_{-1/2}^{(3)}(Z(q); q^2) = 0, \quad Z(q) = \sqrt{2(1-q)/q},$$

if and only if 1 is an eigenvalue of $\mathcal{T}(q)$, that is $\lambda_k(q) = 1$ for some k . That k is unique, the eigenvalue is simple, its eigenvector is the travel null vector R of Proposition 14, and $\phi = D_e^{-1/2} R$ is the corresponding eigenvector of $\mathbf{S}$.

Proof. Theorem 3(iii)–(v), with $c_q > 0$ and $Z > 0$, and Proposition 2; uniqueness of k is the simplicity in Theorem 3(iv), and the eigenvector is Theorem 3(ii) with [5, Lemma “the null space is at most one-dimensional”, lem:nullspace]. $\square$

The zero theory used is that of Koelink–Swarttouw [17], at $\nu = -\frac{1}{2}$ and base q^2 , where its hypothesis $\nu > -1$ holds. The eigenvalues also agree with the Jacobi-matrix description of the zeros of $J_\nu^{(3)}$ by Štampach and Štovíček [28], in its Friedrichs extension; the positive form of $\mathbf{S}$ is the Friedrichs realisation, on which $u_s = (KR)_s \rightarrow 0$. At $q_1, \dots, q_{10}$, located at 90-digit precision, $|J_{-1/2}^{(3)}(Z; q^2)| \leq 2 \cdot 10^{-79}$, and exactly $m-1$ zeros of $\cos(\cdot; q_m^2)$ lie below $Z(q_m)$ (a numerical check, not a certificate).

Proposition 6 (infinitely many travel poles, without T_2). (a) $\lambda_1(q) > (1-q^2)/(2q)$ on $(0, 1)$. In particular every $\lambda_k(q) > 1$ for $q \leq \sqrt{2}-1$, and there is no travel pole in $(0, \sqrt{2}-1]$.

(b) For each fixed k , $\lambda_k(q)/(1-q) \rightarrow (2k-1)^2 \pi^2/8$ as $q \rightarrow 1^-$.

(c) Each λ_k is continuous on $(0, 1)$, and there is $c_k \in (\sqrt{2}-1, 1)$ with $\lambda_k(c_k) = 1$. The c_k are pairwise distinct travel poles; in particular there are infinitely many travel poles.

None of this uses T_2 , [5, Lemma “absolute bound on T_2 ”, lem:T2abs], or a single-crossing property.

Proof. (a) The largest eigenvalue of the positive trace-class $\mathbf{S}$ is strictly less than its trace, since $\mathbf{S}$ has infinitely many positive eigenvalues: $1/\lambda_1 < 2q/(1-q^2)$. And $(1-q^2)/(2q) \geq 1$ exactly for $q \leq \sqrt{2}-1$. At such q no λ_k equals 1, so $1 - \Sigma_1 = F(1, q) \neq 0$ by Theorem 3(iv).

(b) Put $\varepsilon = 1-q$ and $\lambda = \varepsilon y$. The k -th term of $F(\varepsilon y, q)$ is $t_k(y, \varepsilon) = (-1)^k (2y\varepsilon^2)^k q^{k^2} / (q; q)_{2k}$. By the arithmetic–geometric mean inequality, $1 - q^n = \varepsilon \sum_{i < n} q^i \geq n\varepsilon q^{(n-1)/2}$, so $(q; q)_{2k} \geq \varepsilon^{2k} (2k)! q^{k(2k-1)/2}$ and

$$|t_k(y, \varepsilon)| \leq \frac{(2|y|)^k q^{k/2}}{(2k)!} \leq \frac{(2|y|)^k}{(2k-1)!!},$$

uniformly in $\varepsilon \in (0, 1)$. Each $t_k(y, \varepsilon)$ tends to $(-1)^k(2y)^k/(2k)!$ as $\varepsilon \rightarrow 0$, because $q^{k^2} \rightarrow 1$ and $(q; q)_{2k}/\varepsilon^{2k} \rightarrow (2k)!$. The domination is summable locally uniformly in $y \in \mathbb{C}$, so $F(\varepsilon y, 1 - \varepsilon) \rightarrow \cos \sqrt{2}y$ uniformly on compact sets. The limit has the simple zeros $y_j = (2j - 1)^2\pi^2/8$, and the zeros of $y \mapsto F(\varepsilon y, 1 - \varepsilon)$ are the $\lambda_j(q)/\varepsilon$, positive and ordered. By Hurwitz's theorem, for ε small each disc $|y - y_j| < \delta$, $j \leq k$, contains exactly one of them and $|y| \leq y_k + \delta$ contains no other; hence $\lambda_j(q)/\varepsilon \rightarrow y_j$ for $j \leq k$.

(c) $q \mapsto \mathbf{S}(q)$ is continuous in trace norm, hence in operator norm, on $(0, 1)$: its entries are continuous and $\sum_{s, s'} |\mathbf{S}[s, s']|$ converges locally uniformly. The ordered eigenvalues of positive compact operators are 1-Lipschitz in operator norm (min-max), so each $1/\lambda_k$, and so each λ_k , is continuous. By (a) $\lambda_k > 1$ on $(0, \sqrt{2} - 1]$, and by (b) $\lambda_k(q) \rightarrow 0$ as $q \rightarrow 1$, so the intermediate value theorem gives c_k ; it is a travel pole by Corollary 5. If $c_j = c_k$ with $j \neq k$, then $\lambda_j(c_k) = \lambda_k(c_k) = 1$, against simplicity (Theorem 3(iv)).

The same argument with $\lambda_k(q)/q$ in place of $\lambda_k(q)$ gives [5, Lemma “infinitude of the zeros of the bulk block”, lem:infzeros] without T_2 too: $S_e = F(q, q) = \prod_k (1 - q/\lambda_k(q))$ vanishes exactly where some $\lambda_k(q) = q$; $\lambda_k(q)/q > (1 - q^2)/(2q^2) > 1$ for $q < 1/\sqrt{3}$ by (a); $\lambda_k(q)/q \rightarrow 0$ by (b); and two branches cannot both equal q at the same point. $\square$

3.3 Local finiteness, and accumulation at $q = 1$

Lemma 7 (The travel poles are locally finite). *For every $\tau_\star > 0$ there are only finitely many travel poles with $\tau_m > \tau_\star$.*

Proof. By Proposition 2 the travel poles are the zeros in $q \in (0, 1)$ of $g(q) := \cos(Z; q^2) = C(Z^2)$ with $Z^2 = 2(1 - q)/q$, i.e. of

$$g(q) = \sum_{j \geq 0} (-1)^j \frac{2^j q^{j^2} (1 - q)^j}{(q; q)_{2j}}.$$

First, g has no zero near $q = 0$. For $0 < q \leq \frac{1}{10}$ one has $(q; q)_{2j} \geq \prod_{i \geq 1} (1 - q^i) \geq 1 - \frac{q}{1 - q} \geq \frac{8}{9}$ and $(1 - q)^j \leq 1$, so $|g(q) - 1| \leq \frac{9}{8} \sum_{j \geq 1} 2^j q^{j^2} \leq \frac{9}{8} (2q + 4q^4/(1 - q)) \leq 0.24 < 1$, whence $g(q) > 0$ there.

Second, g is real-analytic on $(0, 1)$: the series converges locally uniformly, since on $q \leq 1 - \delta$ the j -th term is bounded by $2^j q^{j^2}/(q; q)_\infty$ and q^{j^2} dominates 2^j . Hence the zeros of g are isolated in $(0, 1)$, and any zero set with an accumulation point in the *open* interval would force $g \equiv 0$, contradicting $g(\frac{1}{10}) > 0$.

Now fix $\tau_\star > 0$. The poles with $\tau_m > \tau_\star$ lie in $q \in [\frac{1}{10}, e^{-\tau_\star}]$, a compact subinterval of $(0, 1)$. An analytic function on a domain whose zeros are isolated has only finitely many zeros on any compact subset; hence there are finitely many such poles. $\square$

Remark 8. Only local finiteness is proved here; that the poles *do* accumulate at $q = 1$ (equivalently $\tau_m \downarrow 0$) needs in addition that there are infinitely many of them, which is Proposition 6(c) (see the paragraph after this remark), or, localised in windows of w , [5, Lemma “infinitude and accumulation of the travel poles”, lem:infpoles]. The present lemma says merely that they cannot accumulate anywhere inside $(0, 1)$, so that any threshold $\tau_\star$ leaves a finite set to be checked directly. Its proof also shows $g > 0$ on $(0, \frac{1}{10}]$, so every travel pole has $q_m > \frac{1}{10}$. Besides the accumulation statement used in Theorems 17, 18 and 20, it makes the splice in [5, Proposition prop:selower] well posed and the exceptional set of [5, Theorem “the amplitude estimate ($\star$)”, thm:star] finite, and it gives the accumulation clause of [5, Lemma lem:infpoles].

With Proposition 6(c), Lemma 7 gives: *the travel poles form an infinite subset of $(\frac{1}{10}, 1)$ whose only accumulation point is $q = 1$; in particular, for every $\tau_\star > 0$ all but finitely many travel poles have*

$\tau_m \leq \tau_*$. This is the form in which the pole set is used below. It needs no estimate on T_2 and no numerical input.

3.4 The numerators, and why they do not vanish at the poles

Proposition 2 identifies the two *denominators*. The two numerators are the companion q -sine at the same two arguments, and that is what settles their non-vanishing at the travel poles. Throughout, $\mathfrak{s}(u) = q^{-1/2} \sin(q^{1/2}u; q^2)$ is the half-step q -sine of [5, Lemma “the gate block in closed form”, lem:P12closed], $\sin(z; q^2) = \sum_{j \geq 0} (-1)^j q^{j^2+j} z^{2j+1} / (q; q)_{2j+1}$.

Lemma 9 (the numerators in closed form). *Identically in $0 < q < 1$,*

$$\Sigma_0 = \frac{2q}{Z} \mathfrak{s}(Z), \quad S_0 = \frac{2q}{z_0} \mathfrak{s}(z_0).$$

With Proposition 2 this identifies all four blocks as one q -trigonometric pair sampled at two arguments a half q -step apart: $(\Sigma_0, 1 - \Sigma_1)$ is $(\frac{2q}{u} \mathfrak{s}(u), \cos(u; q^2))$ at $u = Z$, and (S_0, S_e) is the same pair at $u = z_0 = \sqrt{q} Z$. In particular $S_0 = \frac{2q}{1-q} S_o$ with S_o the odd bulk block of [5, Lemma “the gate block in closed form”, lem:P12closed], which is the second identity of the dictionary [5, (eq:dict)] read as a statement about the explicit series, and is therefore independent of the transfer model.

Proof. Collapse the k -recursion (2) on the even ladders, exactly as Proposition 2 does on the odd ones. From $\alpha_k = 2q^{k+1}/(1 - q^{k+1})$,

$$|\gamma_k| = \alpha_k - \alpha_{k+1} = \frac{2(1-q)q^{k+1}}{(1-q^{k+1})(1-q^{k+2})}, \quad |C_k| = \frac{2(1-q)q^{k+2}}{(1-q^{k+1})(1-q^{k+2})},$$

γ_k and C_k themselves being negative on $(0, 1)$, so both even ladders alternate. For the bulk one, $\sum_{i < j} (2i+1) = j^2$ and the denominators telescope to $(q; q)_{2j}$, giving $\prod_{i < j} |\gamma_{2i}| = [2(1-q)]^j q^{j^2} / (q; q)_{2j}$ and hence

$$S_0 = \sum_{j \geq 0} (-1)^j \alpha_{2j} \prod_{i < j} |\gamma_{2i}| = 2 \sum_{j \geq 0} (-1)^j \frac{[2(1-q)]^j q^{(j+1)^2}}{(q; q)_{2j+1}}.$$

Write $q^{(j+1)^2} = q^{j^2+j} q^j$ and $[2(1-q)]^j q^j = (q^{1/2} z_0)^{2j}$: the sum is $(2q/(q^{1/2} z_0)) \sin(q^{1/2} z_0; q^2) = (2q/z_0) \mathfrak{s}(z_0)$. For the travel ladder, $\sum_{i < j} (2i+2) = j^2 + j$ and the denominators telescope to $(q; q)_{2j}$ again, so $\prod_{i < j} |C_{2i}| = [2(1-q)]^j q^{j^2+j} / (q; q)_{2j}$ and, with $A_{2j} = 2q/(1 - q^{2j+1})$,

$$\Sigma_0 = 2q \sum_{j \geq 0} (-1)^j \frac{[2(1-q)]^j q^{j^2+j}}{(q; q)_{2j+1}} = \frac{2q}{z_0} \sin(z_0; q^2) = \frac{2q^{3/2}}{z_0} \mathfrak{s}(Z) = \frac{2q}{Z} \mathfrak{s}(Z),$$

using $[2(1-q)]^j = z_0^{2j}$ and $z_0 = q^{1/2} Z$. The S_o statement is the identity $S_o = \frac{z_0}{2} \mathfrak{s}(z_0)$ of [5, Lemma “the gate block in closed form”, lem:P12closed] combined with $z_0^2 = 2(1-q)$. Both collapses are the exponent bookkeeping machine-checked in `QTrigIdentities.lean` and are verified numerically off the poles, hence as identities, in `qnumerator_certificate.py`. $\square$

Lemma 10 (half-step rules and the pole invariant). *Identically in $0 < q < 1$, with $c(u) = \cos(u; q^2)$:*

- (i) $c(qu) = c(u) + q^{3/2} u \mathfrak{s}(q^{1/2} u)$;
- (ii) $\mathfrak{s}(qu) = \mathfrak{s}(u) - u c(q^{1/2} u)$;
- (iii) $c(u) c(q^{1/2} u) + q^{3/2} \mathfrak{s}(u) \mathfrak{s}(q^{1/2} u) = 1$.

Consequently, at every travel pole, where $c(Z) = 0$:

$$q^{3/2} \mathfrak{s}(z_0) \mathfrak{s}(Z) = 1,$$

so neither $\mathfrak{s}(Z)$ nor $\mathfrak{s}(z_0)$ vanishes there.

Proof. (i) and (ii) are termwise. From $(q; q)_{2j} = (q; q)_{2j-1}(1 - q^{2j})$,

$$c(u) - c(qu) = \sum_{j \geq 1} (-1)^j q^{j^2+j} u^{2j} / (q; q)_{2j-1};$$

shifting $j \mapsto j+1$ and factoring out $-qu$ leaves $-qu \sin(qu; q^2) = -q^{3/2} u \mathfrak{s}(q^{1/2} u)$, which is (i). From $(q; q)_{2j+1} = (q; q)_{2j}(1 - q^{2j+1})$, $\mathfrak{s}(u) - \mathfrak{s}(qu) = q^{-1/2} \sum_{j \geq 0} (-1)^j q^{j^2+j} (q^{1/2} u)^{2j+1} / (q; q)_{2j}$; factoring out $q^{1/2} u$ leaves $u c(q^{1/2} u)$, which is (ii). This is the exponent matching machine-checked as `cshift_exponent` and `sshift_exponent`. For (iii), substitute (i) and (ii) into the Hahn–Exton q -Wronskian $\mathfrak{s}(u)c(qu) - c(u)\mathfrak{s}(qu) = u$ ([31, eq. (4.3.8)]; [22, eq. (4.21)]): the $\mathfrak{s}(u)c(u)$ terms cancel and what is left is $u[c(u)c(q^{1/2} u) + q^{3/2} \mathfrak{s}(u)\mathfrak{s}(q^{1/2} u)]$, so the q -Wronskian and (iii) are equivalent for $u \neq 0$. Identity (iii) is the q -Pythagorean identity of [18]. Taking $u = Z$ and $c(Z) = 0$ gives the pole invariant, which is the step already used inside the proof of [5, Proposition prop:P12second]. $\square$

Proposition 11 (the numerators do not vanish at the travel poles). *At every travel pole q_m ,*

$$\Sigma_0(q_m) S_0(q_m) = \frac{2q_m}{1 - q_m} \neq 0. \quad (3)$$

In particular $\Sigma_0(q_m) \neq 0$ and $S_0(q_m) \neq 0$, at every travel pole, with no asymptotic input and no exceptional set. Moreover $|\Sigma_0(q_m)| |S_0(q_m)| \geq q_m w_m^2$, so at least one of the two numerators exceeds $\sqrt{q_m} w_m$ in modulus.

Proof. By Lemma 9, $\Sigma_0 S_0 = (4q^2/(z_0 Z)) \mathfrak{s}(Z) \mathfrak{s}(z_0)$, and by Lemma 10 the product $\mathfrak{s}(Z) \mathfrak{s}(z_0)$ equals $q^{-3/2}$ at a travel pole. Since $z_0 Z = z_0^2 / \sqrt{q} = 2(1 - q) / \sqrt{q}$, the two powers of q combine to $4q^2 \cdot q^{-3/2} \cdot \sqrt{q} / (2(1 - q)) = 2q / (1 - q)$. The size statement follows from $1 - q \leq \tau = 2/w^2$. $\square$

4 The transfer model and the local form at a travel pole

This section connects the explicit operators to the growth series. We first collect the objects the transfer model is phrased in; their construction, and every identity between them that is used below, is proved in [5, §sec:assembly].

4.1 Notation

Throughout, $|q| < 1$, $x = q^{1/2}$, and at a travel pole q_m we write $\tau_m = -\log q_m$, $w_m = \sqrt{2/\tau_m}$ and $x_m = \sqrt{q_m}$.

The two operators. On $\ell^1(\mathbb{Z}_{\geq 0})$ the *travel operator* is $\mathcal{T} = D_e K$ with $K = [q^{\max(s, s')}]_{s, s' \geq 0}$ and $D_e = \text{diag}(e_s)$, $e_s = 2q^{1+s}$; on $\ell^1(\mathbb{Z}_{\geq 1})$ the *bulk operator* is $M_0[b, a] = 2q^b q^{\max(a, b)}$. These are the travel and bulk instances of the section identity of [5, Definition “section identity”, def:model]; both are compact, and $(I - \mathcal{T})^{-1}$ and $R_0 = (I - M_0)^{-1}$ are meromorphic on $|q| < 1$ with poles only where $1 - \Sigma_1 = 0$, respectively $S_e = 0$ [5, Corollary “singularities of the two gapless resolvents”, cor:sing]. Put $E_b = 2q^b$, $u_b = 2q^{2b}$, $v_a = q^a$ and

$$t_1 = v^\top R_0 u,$$

an explicit bulk quantity; one has $t_1 = P_{12}/S_e$ identically, where P_{12} is the off-diagonal entry of the bulk cocycle, a Hahn–Exton function $J_{3/2}^{(3)}$ in closed form [5, Corollary “the cocycle is the same object”, cor:cocycle].

The gap term. With $\mathfrak{g} = \mathfrak{g}(q, y) = \sum_{L \geq 1} q^L y^{L-1} = q/(1 - qy)$, the gap-marked bulk system is

$$K_{\mathfrak{g}}(a, b) = q^{\max(a, b)} + q^{a+b} \mathfrak{g}, \quad P_b = 2q^b \left(q^b + \sum_{a \geq 1} K_{\mathfrak{g}}(a, b) P_a \right) \quad (b \geq 1). \quad (4)$$

The gap term is of rank one, $2q^b \cdot q^{a+b} = u_b v_a$, and by the Sherman–Morrison formula $P = B_0 R_0 u$ with

$$B_0 = 1 + \mathfrak{g} \sum_{b \geq 1} q^b P_b = \frac{1}{1 - \mathfrak{g} t_1}, \quad B_{0z} = 1 + y \mathfrak{g} \sum_{b \geq 1} q^b P_b$$

[5, Proposition “Möbius factorisation”, prop:mobius; Remark “the bulk sum of the model”, rem:bulksources]. The two evaluations of the gap kernel are $\mathfrak{g}_V = \mathfrak{g}(q, 1) = \frac{q}{1-q}$ and $\mathfrak{g}_U = \mathfrak{g}(q, q) = \frac{q}{1-q^2}$.

The junction vectors. For $s \geq 0$,

$$\mu_s^{(y)} = B_0 x^{2s+1} + \sum_{b \geq 1} \frac{P_b}{2} \left(x^{\max(2b-1, 2s+1)} + x^{\max(2b+1, 2s+1)} \right), \quad (5)$$

$$E_s^{(y)} = B_0^{[s]} x^{2s} + B_0 x^{2s+2} + \sum_{b \geq 1} P_b \left(x^{\max(2s, 2b)} + x^{\max(2s+2, 2b)} \right), \quad B_0^{[0]} = B_{0z}, \quad B_0^{[s]} = B_0 \quad (s \geq 1), \quad (6)$$

and $\lambda^{(y)} = E^{(y)} + 2\mu^{(y)}$. These are the near- and far-junction vectors of the marker sites; the site costs they encode are computed in [5, Corollary “the marker junctions”, cor:marker], and the rank of the junction is discussed in [5, Proposition “the marker junction is not rank one”, prop:junction].

The bivariate series. With $c(g)$ the cycle count and $\ell_R(g)$ the relaxed length,

$$\mathcal{W}(x, y) = \sum_{n, c \geq 0} w_{n, c} x^n y^c, \quad w_{n, c} = \#\{g : \ell_R(g) = n, \quad c(g) = c\},$$

so that $V(x) = \mathcal{W}(x, 1)$ and, since $\ell_T = \ell_R + 2c$ for every element (formalised in Lean, `metricAll`), $U(x) = \mathcal{W}(x, x^2) = \mathcal{W}(x, q)$ [5, §sec:bridge].

The gate. Write $s = \mathfrak{g}_V t_1 = \frac{q}{1-q} t_1$ and $b_0 = S_0/(1 - S_1)$. The gate is

$$b_0 > 0 \quad \text{and} \quad s < 1; \quad (7)$$

it holds at every travel pole, with $|s| \leq 0.983$ for $\tau_m \leq 5 \cdot 10^{-3}$ and by direct evaluation at the six poles above that threshold, whose enumeration has the standing stated in [5, Theorem “the amplitude estimate ($\star$)”, thm:star] (see [5, Corollary “the gate closes”, app:gate]). The half $|s| < 1$ holds unconditionally, and the half $b_0 > 0$ through the closed form of b_0 , which rests on the transfer model.

4.2 The model (M)

The strand-walk input that identifies the operators above with the growth series is stated once and named.

Theorem 12 (the transfer model (M)). (M1) Local cost law. *The site cost between adjacent active edges carrying deposits d, d' is $x^{\max(|d|, |d'|)}$, independent of the crossing counts and of the travel indicator. Bulk deposits are even, $|d| = 2s$, and travel deposits odd, $|d| = 2s + 1$, so this cost is $q^{\max(s, s')}$ in the bulk and $x^{2\max(s, s') + 1}$ on the travel side, and the edge weights are $2q^s$ and $2x^{2s+1}$ respectively: [5, Definition “section identity”, def:model] in its two instances. At the near marker site the cost is $x^{\text{Site}_0(d_L, d_R)}$ with $\text{Site}_0(d_L, d_R) = \max(|d_L - 1|, |d_R|)$, d_L the last bulk deposit and d_R the first travel deposit, for all four marker data $(\varepsilon^*, \delta^*)$; at the far marker site it is the $(\varepsilon^*, \delta^*)$ -dependent cost of [5, Corollary “the marker junctions”, cor:marker].* (M2) Shield law. *A maximal interior run of L forced gap edges costs q^L and carries $L - 1$ isolated cycles; hence the gap-bridge kernel is $\mathfrak{g}(q, y) = q/(1 - qy)$ and the gap-marked bulk kernel is the $K_{\mathfrak{g}}$ of (4).* (M3) Decomposition. *The bivariate series of [5, §sec:bridge] is $\mathcal{W} = \mathcal{W}_+ + \mathcal{W}_- + \mathcal{W}_0$, the sectors $k^* > 0$, $k^* < 0$, $k^* = 0$, with*

$$\begin{aligned} \mathcal{W}_+ &= x^{-1} \langle \mu^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle, & \mathcal{W}_- &= x^{-1} \langle \tfrac{1}{2} E^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle, \\ \mathcal{W}_+ + \mathcal{W}_- &= \frac{1}{2x} \langle \lambda^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle, \end{aligned} \tag{8}$$

where $\mathcal{T} = D_e K$ is the travel operator of [5, Definition “section identity”, def:model], $D_e = \text{diag}(e_s)$ with $e_s = 2q^{1+s}$, and $\mu^{(y)}, E^{(y)}, \lambda^{(y)}$ are as in (5)–(6). The sum over the marker data $(\varepsilon^*, \delta^*)$ is already carried out in these vectors: $\delta^* = 1$ at the far site gives 2μ and $\delta^* = 0$ gives E when $k^* > 0$, and the roles are exchanged when $k^* < 0$. The $k^* = 0$ sector is $\mathcal{W}_0 = A_0 + B_0 A_1 + B_0^2 A_2$, with A_i built from bulk quantities alone; it contains no travel resolvent.

Proof and standing. Let $(k^*, \varepsilon^*, \delta^*, d)$ parametrise the elements of W_{univ} as in [3, Theorem “Normal form and lamp model”, thm:normal-form]. Two facts are formalised in Lean 4 with the standard axioms only:

- the affine realisation intertwines the three generators and is injective, and every element of the model is reachable, so the word length of the model is the Cayley length in W_{univ} (RJPhi.lean: `intertwine_s1, intertwine_s2, intertwine_s3, injective; reachability is reaches_of_phiZ` in PhilLipschitz.lean);
- $\text{wordLength} = \ell_R + 2c$ for every element, with ℓ_R and c the site-local closed forms `1RTrue, cTrue` of `CorrectedSpan.lean` (RJMetricAll.lean, theorem `metricAll`).

Given these, ℓ_R and c are sums of site and edge costs whose only non-bulk sites are 0 and k^* ([5, Corollaries `cor:localcost` and `cor:marker`]; a gap run of length L contributes $q^L y^{L-1}$ and a travel-interior site is never a cut). Summing these site costs chain by chain gives (4) with source u , the vectors (5)–(6), and the three sectors of (8); this is [5, Theorem “(M3’)”, thm:M3prime], proved in [5, Appendix `app:M3prime`], and the regrouping of $\mathcal{W}_0$ is [5, Remark “the regrouping of $\mathcal{W}_0$ ”, M3:rem:W0regroup]. That last step is a hand proof, with two independent adversarial reviews, and it is not formalised. Every identity holds in $\mathbb{Z}[y][[x]]$, and the series agree with enumeration as recorded in [5, §sec:bridge]. $\square$

One caveat: ℓ_R in $\mathcal{W}$ is the closed-form relaxed length of [3]. That it also equals the minimum over relaxed realisations ([3, Remark `rem:metric-formula-status`]) affects only the *interpretation* of V as a relaxed count, not U and not the series identities. How (M) was carried in earlier versions, and which of its weak links the proof removes, is recorded in [5, §sec:inputL].

$V = \mathcal{W}(x, 1)$ and $U = \mathcal{W}(x, q)$ ([5, §sec:bridge]), the two evaluations differing only through $\mathfrak{g}_V = \frac{q}{1-q}$, $\mathfrak{g}_U = \frac{q}{1-q^2}$ and the slot B_{0z} ; and $\lambda^{(y)}, \mu^{(y)}, \mathcal{W}_0$ are holomorphic at a travel pole q_m , by [5,

(eq:smresolvent)], as soon as $S_e(q_m) \neq 0$ and $1 - \mathfrak{g}t_1 \neq 0$ there, which is [5, Proposition prop:selower] and [5, Corollary “the gate closes”, app:gate] ($|\mathfrak{g}_V t_1| \leq 0.983$, and $|\mathfrak{g}_U t_1| < |\mathfrak{g}_V t_1|$). In particular $\mathcal{W}_0$ has no travel pole: by [5, Remark “the regrouping of $\mathcal{W}_0$ ”, M3:rem:W0regroup] it is a polynomial in B_0 with coefficients built from the bulk vector P and monomials, so its only possible singularities are at $S_e = 0$ and at $1 - \mathfrak{g}t_1 = 0$.

The growth series are defined as power series, with radius of convergence $x_1 = \sqrt{q_1} = 1/\beta_2 = 0.6704\dots$ (Corollary 19), and the local forms below are statements about functions on a larger disc. The following lemma supplies that continuation and identifies it with the series.

Lemma 13 (meromorphic continuation). *Let y be either a complex constant with $|y| \leq 1$ or $y = x^2$. Then the specialisation $x \mapsto \mathcal{W}(x, y)$ of the series of [5, Theorem “(M3’)”, thm:M3prime] converges near $x = 0$ and is the Taylor series at 0 of a single-valued meromorphic function on the disc $|x| < 1$, given there by (8) and $\mathcal{W}_0 = A_0 + B_0 A_1 + B_0^2 A_2$ read as analytic expressions. Its poles in $0 < |x| < 1$ lie over points $q = x^2$ at which $1 - \Sigma_1(q) = 0$, $S_e(q) = 0$ or $1 - \mathfrak{g}(q, y) t_1(q) = 0$. In particular $V = \mathcal{W}(x, 1)$ and $U = \mathcal{W}(x, x^2)$ are meromorphic on $|x| < 1$, and $G_0^T = \Sigma_0/(1 - \Sigma_1)$ is meromorphic on $|q| < 1$. (The continuation concerns the explicit operators of [5, §§sec:sect–sec:mob]; its identification with U and V is Theorem 12.)*

Proof. Put $q = x^2$, so $|q| < 1$; $\mathfrak{g} = q/(1 - qy)$ is holomorphic on the disc, since $|qy| < 1$ in both cases.

Operators. $q \mapsto \mathcal{T}(q)$ and $q \mapsto M_0(q)$ are holomorphic on $|q| < 1$ as maps into the bounded operators on ℓ^1 , in operator norm: the row truncations M_N in the proof of [5, Lemma “compactness and meromorphy”, lem:fred] have finitely many non-zero rows, each a norm-convergent ℓ^∞ -valued power series in q (the row s is $e_s \sum_{s'} q^{\max(s, s')} \delta_{s'}$), hence are holomorphic in norm, and, with M either operator and $\vartheta \in \{0, 1\}$ its exponent offset in [5, Definition “section identity”, def:model], $\|M - M_N\| \leq \sum_{s > N} 2|q|^{\vartheta+s} \rightarrow 0$ uniformly on $|q| \leq r < 1$; a locally uniform norm limit of holomorphic operator-valued maps is holomorphic. With compactness ([5, Lemma “compactness and meromorphy”, lem:fred]) and invertibility near $q = 0$ ([5, Lemma “existence, uniqueness, analyticity”, lem:modelexist]), the analytic Fredholm theorem makes $(I - \mathcal{T})^{-1}$ and $R_0 = (I - M_0)^{-1}$ meromorphic on $|q| < 1$ in operator norm, with poles only where $1 - \Sigma_1 = 0$, resp. $S_e = 0$ ([5, Corollary “singularities of the two gapless resolvents”, cor:sing]).

Bulk quantities. $u = (2q^{2b})_b$ is a norm-convergent, hence holomorphic, ℓ^1 -valued function, so $\Psi = R_0 u$ is meromorphic ℓ^1 -valued and $t_1 = v^\top \Psi$ is meromorphic. $1 - \mathfrak{g}t_1$ tends to 1 as $x \rightarrow 0$, so it is not identically zero, and $B_0 = 1/(1 - \mathfrak{g}t_1)$, $P = B_0 \Psi$, $\mathfrak{g}S = B_0 - 1$ and $B_{0z} = 1 + y(B_0 - 1)$ are meromorphic ([5, Remark “the regrouping of $\mathcal{W}_0$ ”, M3:rem:W0regroup]).

Boundary vectors and the sectors $k^ \neq 0$.* Every entry of [5, (eq:app-mu)–(eq:app-lambda)] is a combination of B_0 , B_{0z} and sums $\sum_b P_b x^{m_b}$ with $m_b \geq 0$, so $|\mu_s^{(y)}|, |E_s^{(y)}|, |\lambda_s^{(y)}| \leq C(|B_0| + |B_{0z}| + \|P\|_{\ell^1})$ uniformly in s , locally uniformly in x away from the poles of B_0 and Ψ . Hence $D_e \lambda^{(y)}$ has ℓ^1 -tails $\sum_{s > N} |e_s \lambda_s^{(y)}| = O(|q|^{N+2})$, locally uniformly, and is meromorphic ℓ^1 -valued; so is $(I - \mathcal{T})^{-1} D_e \lambda^{(y)}$. Its pairing with the bounded vectors $\mu^{(y)}$ and $\frac{1}{2} E^{(y)}$ is an absolutely convergent sum, locally uniformly convergent away from the poles (an ℓ^1 -valued continuous map has uniformly small tails on compact sets), hence meromorphic. This gives $\mathcal{W}_\pm$; the prefactor x^{-1} is at most a pole at $x = 0$, and is removed below.

The sector $k^ = 0$.* By [5, Remark “the regrouping of $\mathcal{W}_0$ ”, M3:rem:W0regroup], $\mathcal{W}_0 = A_0 + B_0 A_1 + B_0^2 A_2$. The A_i are not finite sums: each is an absolutely convergent double sum, over the left and the right chain kinds $\mathbf{E}, \mathbf{Z}, \mathbf{N}_{\pm, b}$ of [5, (eq:app-W0)] ($b \geq 1$), of monomials $x^j y^\chi$ with $j \geq 0$, $\chi \in \{0, 1\}$, times at most two entries of Ψ . Since $|x^j y^\chi| \leq 1$, the double sum is dominated by a constant times $(1 + \|\Psi\|_{\ell^1})^2$, converges locally uniformly where Ψ is holomorphic, and is meromorphic.

Identification with the x -adic series. Choose $r_0 > 0$ so small that on $|x| < r_0$ the operators $\mathcal{T}$, M_0 and $M_0 + \mathbf{g}uv^\top$ have ℓ^1 -norm below $\frac{1}{2}$ ([5, Lemma “existence, uniqueness, analyticity”, lem:modeexist]); the rank-one term has norm at most $|\mathbf{g}| \|u\|_{\ell^1}$ and $|\mathbf{g}t_1| < \frac{1}{2}$. There every object above is the sum of a norm-convergent Neumann series $((I - \mathcal{T})^{-1}, R_0, P = (I - M_0 - \mathbf{g}uv^\top)^{-1}u$, and $B_0 = \sum_n (\mathbf{g}t_1)^n$), whose terms are holomorphic on $|x| < r_0$ and whose Taylor series at 0 are the terms of the x -adic Neumann series of [5, Lemma M3:lem:conv], of x -adic order tending to infinity. If $\sum_n h_n$ converges uniformly on a disc about 0, the Taylor coefficients of the sum are the sums of the Taylor coefficients of the h_n (Weierstrass); when $\text{ord } h_n \rightarrow \infty$ each of these sums is finite and is the x -adic sum. Applying this to each Neumann series, to the absolutely convergent pairings and to the double sums A_i , the Taylor series at 0 of the analytic $\mathcal{W}(x, y)$ is the x -adic series of [5, Theorem “(M3’)”, thm:M3prime], specialised at y (the specialisation $\mathbb{Z}[y][[x]] \rightarrow \mathbb{C}[[x]]$ at a constant y , or at $y = x^2$, is x -adically continuous and commutes with the sums). In particular that function is holomorphic at $x = 0$. At $y = 1$ the series is V , and at $y = x^2$ it is U ([5, Corollary cor:M3prime-growth]). A power series that agrees near 0 with a function meromorphic on the disc is continued by it, uniquely. For G_0^T , Σ_0 and $1 - \Sigma_1 \neq 0$ are holomorphic on $|q| < 1$ ([5, Theorem thm:blocks], first clause). $\square$

Proposition 14 (the shape of the local form). *Let q_m be a travel pole, let $p \geq 1$ be the order of the zero of $1 - \Sigma_1$ there (finite, Lemma 7), and let π be the order of the pole of $(I - \mathcal{T})^{-1}$ there (finite, [5, Lemma “compactness and meromorphy”, lem:fred]). Then $\pi \geq p \geq 1$ and, for each y ,*

$$\mathcal{W}(x, y) = \frac{\mathcal{N}_y}{(q - q_m)^\pi} + (\text{poles of order } < \pi) + H_m, \quad H_m \text{ holomorphic at } q_m,$$

with the leading numerator a perfect square

$$\mathcal{N}_y = \frac{c_\star}{2x_m} \Pi_y(q_m)^2, \quad \Pi_y(q_m) := \langle \lambda^{(y)}, R \rangle, \quad c_\star \neq 0, \quad x_m = \sqrt{q_m}, \quad (9)$$

where R spans $\ker(I - \mathcal{T}(q_m))$, normalised by its sections,

$$\sum_{s \geq 0} q^{ks} R_s = \Sigma_k(q_m) \quad (k \geq 0), \quad \text{in particular } \sum_s R_s = \Sigma_0(q_m), \quad \sum_s q^s R_s = \Sigma_1(q_m) = 1,$$

and L spans $\ker(I - \mathcal{T}(q_m)^\top)$, explicitly $L_s = R_s/e_s$. There are no cross terms between the sectors. In particular $\Pi_y(q_m) \neq 0$ implies that q_m is a pole of $\mathcal{W}(\cdot, y)$, of order exactly π .

Proof. Write $\mathcal{R}(q) = (I - \mathcal{T}(q))^{-1} = \sum_{j=1}^\pi A_j(q - q_m)^{-j} + \text{hol}$, with $A_\pi \neq 0$. Multiplying $(I - \mathcal{T}(q))\mathcal{R}(q) = I$ by $(q - q_m)^\pi$ and letting $q \rightarrow q_m$ gives $(I - \mathcal{T}(q_m))A_\pi = 0$, so $\text{ran } A_\pi \subseteq \ker(I - \mathcal{T}(q_m))$, which is one-dimensional by [5, Lemma “the null space is at most one-dimensional”, lem:nullspace]; the same step on $\mathcal{R}(q)(I - \mathcal{T}(q)) = I$ gives $A_\pi(I - \mathcal{T}(q_m)) = 0$. Hence $A_\pi x = c_\star \langle L, x \rangle R$ with $c_\star \neq 0$, R spanning the kernel and L the cokernel. The normalisation of R is [5, Lemma “the null space is at most one-dimensional”, lem:nullspace]: with the sections $G_k^R = \sum_s q^{ks} R_s$, one has $G_k^R = G_1^R \Sigma_k$ and $\Sigma_1(q_m) = 1$, so $G_1^R = 1$ gives $G_k^R = \Sigma_k(q_m)$. That $L_s = R_s/e_s$ spans the cokernel follows from $\mathcal{T} = D_e K$ with $D_e = \text{diag}(e_s)$ and K symmetric: from $D_e K R = R$ one gets $K D_e (D_e^{-1} R) = K R = D_e^{-1} R$, that is $\mathcal{T}^\top L = L$, and $L \in \ell^\infty$ because $|R_s| \leq |e_s| \|R\|_{\ell^1}$; the cokernel is one-dimensional by Fredholm. Substituting the Laurent expansion into $\mathcal{W}_+ + \mathcal{W}_-$ of (8), and using that $\lambda^{(y)}, \mathcal{W}_0$ are holomorphic at q_m , the leading coefficient is $(2x_m)^{-1} \langle \lambda, A_\pi D_e \lambda \rangle = (2x_m)^{-1} c_\star \langle L, D_e \lambda \rangle \langle \lambda, R \rangle = (2x_m)^{-1} c_\star \langle \lambda, R \rangle^2$, since $D_e L = R$. It is non-zero exactly when $\Pi_y(q_m) \neq 0$, and then q_m is a pole of order π . Finally $\langle \mathbf{1}, \mathcal{R}E \rangle = \Sigma_0/(1 - \Sigma_1)$ ([5, Proposition “the travel k -sum telescopes”, prop:travelsum]) has a pole of order exactly p at q_m , since $\Sigma_0(q_m) \neq 0$ (Proposition 11), while a matrix element of $\mathcal{R}$ has pole order at most π ; hence $\pi \geq p$. $\square$

4.3 The junction pairing (R-J)

(R-J) is the statement that $\Pi_y(q_m) = \langle \lambda^{(y)}, R \rangle \neq 0$ at infinitely many travel poles, for $y = 1$ and $y = q$. Earlier versions carried it as a hypothesis. It is now a theorem.

Theorem 15 (closed form of the junction pairing). *Let q_m be a travel pole, $x = \sqrt{q_m}$, and R normalised as in Proposition 14. Then*

$$(1 - \mathfrak{g}_V t_1) \Pi_1(q_m) = \frac{2q(1+x)}{1-q} \left(t_1 + \frac{1+x}{2x} \right), \quad (10)$$

$$\Pi_q(q_m) = \frac{2q B_0(1+x)}{1-x} \left(t_1 \frac{1-x+q}{1+q} + \frac{1}{2x} \right), \quad B_0 = \frac{1}{1 - \mathfrak{g}_U t_1}. \quad (11)$$

Consequently $\Pi_1(q_m) > 0$ and $\Pi_q(q_m) > 0$ at every travel pole q_m .

Proof. Throughout $q = q_m$.

Reduction to the ungapped bulk vector. Put $\Psi_0 = 1$ and let $\Psi = (\Psi_b)_{b \geq 1} = R_0 u$ be the solution of the bulk system with $\mathfrak{g} = 0$ and source u , that is

$$\Psi_b = 2q^b F_b, \quad F_t := \sum_{b \geq 0} q^{\max(b,t)} \Psi_b \quad (b \geq 1),$$

the term $b = 0$ of F_t being the source q^t . Then $t_1 = v^\top R_0 u = \sum_{b \geq 1} q^b \Psi_b$ and, by [5, (eq:smresolvent)], $P = R_0 u / (1 - \mathfrak{g} t_1) = B_0 \Psi$, where $B_0 = 1 / (1 - \mathfrak{g} t_1)$ ([5, Remark “the bulk sum of the model”, rem:bulksource]). So every bulk weight in (5)–(6) is $B_0 \Psi_b$, with $b = 0$ standing for the empty and gap-opening chains. In the variable $q = x^2$,

$$\begin{aligned} x^{\max(2s,2b)} &= q^{\max(s,b)}, & x^{\max(2s+2,2b)} &= q q^{\max(s,b-1)}, \\ x^{\max(2s+1,2b-1)} &= x q^{\max(s,b-1)}, & x^{\max(2s+1,2b+1)} &= x q^{\max(s,b)}, \end{aligned}$$

with $\max(s, -1) = s$. At $y = 1$ the slot B_{0z} equals B_0 , and (5), (6) combine into

$$\lambda_s^{(1)} = B_0 \sum_{b \geq 0} \Psi_b \left((1+x) q^{\max(s,b)} + (q+x) q^{\max(s,b-1)} \right), \quad (12)$$

hence $\langle \lambda^{(1)}, R \rangle = B_0((1+x)X + (q+x)Y)$ with

$$X = \sum_{b \geq 0} \Psi_b (KR)_b, \quad Y = \sum_{b \geq 0} \Psi_b G_b, \quad (KR)_b = \sum_{t \geq 0} q^{\max(b,t)} R_t, \quad G_b = \sum_{s \geq 0} q^{\max(s,b-1)} R_s.$$

All double sums converge absolutely ($\Psi \in \ell^1$ with $|\Psi_b| = O(q^{2b} |\Psi|_{\ell^1})$, $R \in \ell^1$), so they may be summed in either order.

Two facts about R . From $\mathcal{T}R = R$, $R_s = 2q^{s+1}(KR)_s$ for every $s \geq 0$; and the normalisation $\sum_s q^s R_s = \Sigma_1(q_m) = 1$ of Proposition 14 says $(KR)_0 = G_0 = 1$. In particular $R_0 = 2q$.

The X shift identity. Summing first over t and splitting off $b = 0$, $X = 1 + S$ with $S = \sum_{b \geq 1} \Psi_b (KR)_b$. Summing first over b instead, $X = \sum_t R_t F_t$. The term $t = 0$ is $R_0 F_0 = 2q(1+t_1)$, since $F_0 = \sum_b q^b \Psi_b = 1 + t_1$; for $t \geq 1$, $R_t F_t = 2q^{t+1}(KR)_t \cdot \Psi_t / (2q^t) = q \Psi_t (KR)_t$. Hence $X = 2q(1+t_1) + qS$, and eliminating S ,

$$X = 1 + \frac{X - (1+t_1)R_0}{q}, \quad \text{so} \quad X = \frac{q(1+2t_1)}{1-q}.$$

The Y shift identity. Splitting off $b = 0$ and using $G_b = (KR)_{b-1} = R_{b-1}/(2q^b)$ for $b \geq 1$, $Y = 1 + \sum_{s \geq 0} \Psi_{s+1} R_s / (2q^{s+1})$. Summing first over b instead, and using $q^{\max(s, b-1)} = q^{-1} q^{\max(s+1, b)}$, $Y = \sum_s R_s F_{s+1} / q = \sum_s R_s \Psi_{s+1} / (2q^{s+2})$. The two expressions give

$$Y = 1 + qY, \quad \text{so} \quad Y = \frac{1}{1-q}.$$

The closed forms. With $q = x^2$, $(1+x)X + (q+x)Y = ((1+x)q(1+2t_1) + q+x)/(1-q)$, and since $(1+x)q + q + x = x(1+x)^2$ this is (10). At $y = q$ only the slot $s = 0$ of $E^{(q)}$ changes, from B_0 to $B_{0z} = 1 + q \mathbf{g}_U S = B_0(1 - t_1 q / (1+q))$ (using $\mathbf{g}_U S = B_0 - 1 = \mathbf{g}_U t_1 B_0$). So $\Pi_q = B_0((1+x)X + (q+x)Y) + (B_{0z} - B_0)R_0$, now with $B_0 = 1/(1 - \mathbf{g}_U t_1)$, and collecting terms with $(1+x)^2(1-x+q) = (1+x)(1+q) - q(1-q)$ (both equal $1+x+x^3+x^4$) gives (11).

Positivity for $\tau_m \leq 5 \cdot 10^{-3}$. By [5, Corollary “the gate closes”, app:gate], $|\mathbf{g}_V t_1| \leq 0.983$, and $\mathbf{g}_U < \mathbf{g}_V$. Hence $1 - \mathbf{g}_U t_1 \geq 0.017$ for both y , and $q|t_1| \leq 0.983(1-q)$, so $|t_1| < 0.12$ once $q \geq 0.9$ (here $q \geq e^{-0.005}$). Since $(1+x)/(2x) \geq 1$, $1/(2x) > 1/2$ and $0 < (1-x+q)/(1+q) < 1$, both brackets are positive, and so are the prefactors.

Positivity at the poles above that threshold. This part is a computer-assisted certificate: at $q_1, \dots, q_{12}$, Π_1 and Π_q are enclosed in MPFR interval arithmetic and found positive, together with $1 - \mathbf{g}_U t_1 > 0$ (code/zeta_probe/rj_certificates/r35rust/src/bin/r54b.rs). The pole index comes from code/zeta_probe/rj_certificates/r39pole, an interval-arithmetic zero count of $1 - \Sigma_1$ which isolates exactly 13 zeros on $[0, 0.9988]$ (the right endpoint taken as its binary64 value), each in its own cell. Its tiling of the interval was repaired after a first review found gaps of order 10^{-77} between cells, and the repaired code was rebuilt and re-run from current source by a second reviewer; the trusted base is MPFR and the tool’s own code. $\square$

Only infinitely many poles are needed for Theorems 18 and 20, and the analytic part ($\tau_m \leq 5 \cdot 10^{-3}$) supplies them; the certificate is needed only for the statement “at every travel pole”. The algebraic core is formalised in Lean 4 ([5, §sec:RJlean]); the eigen-relations of R , the gate and the analytic estimates behind it enter the formal statements as hypotheses and are not formalised.

Corollary 16 ((R-J) at the dominant pole). $\Pi_1(q_1) > 0$ and $\Pi_q(q_1) > 0$; this is the case $m = 1$ of Theorem 15.

5 Transcendence of the travel resolvent, of V and of U

The relaxed series is the travel resolvent dressed by the bulk. The resolvent itself is settled unconditionally; passing from it to V uses the transfer model (Theorem 12) and the junction pairing (Theorem 15).

Theorem 17 (the travel resolvent). *The travel resolvent $G_0^T = \Sigma_0 / (1 - \Sigma_1)$ is transcendental over $\mathbb{Q}(x)$. This is unconditional: it uses no growth hypothesis, no asymptotic theorem, and no statement about the strand-walk model. With the pole set taken from Proposition 6, as in the proof below, it has no numerical input. The second route to the pole set, [5, Lemma “infinitude and accumulation of the travel poles”, lem:infpoles], consumes only $|T_2(m\pi)| < 1$ from the contour bound of [5, Lemma “absolute bound on T_2 ”, lem:T2abs], against a computed worst value 0.027; that bound is a verified enclosure over the parameter region, not a sampled one.*

Proof. Σ_0 and $1 - \Sigma_1$ are holomorphic on $|q| < 1$ ([5, Theorem thm:blocks], first clause) and $1 - \Sigma_1 \not\equiv 0$. The zero set of $1 - \Sigma_1$ in $(0, 1)$ is infinite and accumulates exactly at $q = 1$ (Proposition 6(c) together with Lemma 7; alternatively [5, Lemma “infinitude and accumulation of the travel poles”,

lem:infpoles], which consumes from [5, Lemma “absolute bound on T_2 ”, lem:T2abs] only $|T_2| < 1$). At each such q_m the numerator does not vanish, by Proposition 11. Hence every q_m is a pole of G_0^T , of order the vanishing order of $1 - \Sigma_1$ there, which is finite. So G_0^T has infinitely many poles in $|q| < 1$ accumulating at $|q| = 1$; under $q = x^2$ these are infinitely many poles in $|x| < 1$ accumulating at $|x| = 1$. An algebraic function over $\mathbb{Q}(x)$ has only finitely many poles in $|x| < 1$, so no algebraic function has such a pole set; hence G_0^T is transcendental. It is not D -finite either, in q or in x , and $|q| = 1$ is its natural boundary (Theorems 26 and 31): for D -finiteness the finiteness of the pole set of an algebraic function is replaced by Lemma 25, which confines the poles in the disc of a solution of a linear differential equation to the finitely many zeros of its leading coefficient. $\square$

Theorem 18. *The relaxed growth series V is transcendental over $\mathbb{Q}(x)$. (Standing: Theorems 12 and 15.)*

Proof. By Lemma 13, V is meromorphic on $|x| < 1$, and the local forms below are those of this continuation. By Proposition 14, at a travel pole q_m the relaxed series has the local form $V = \mathcal{N}_1(q - q_m)^{-\pi} + (\text{poles of order } < \pi) + H_m$ with H_m holomorphic, $\pi \geq 1$, and $\mathcal{N}_1 = (2x_m)^{-1} c_\star \Pi_1(q_m)^2$ with $c_\star \neq 0$. That proposition consumes two non-vanishing statements, and they are separate: $S_e(q_m) \neq 0$, in the quantitative form $|S_e(q_m)| \geq 0.35\sqrt{\tau_m}$ of [5, Proposition prop:selower], which makes the bulk boundary vectors $\lambda^{(1)}, \mu^{(1)}$ holomorphic at q_m and says nothing about vanishing; and $\Sigma_0(q_m) \neq 0$, from Proposition 11, which pins the pole order $\pi \geq p \geq 1$; and $\Pi_1(q_m) \neq 0$ at every travel pole, by Theorem 15.

So the leading Laurent coefficient is non-zero at every travel pole, and V has a pole at each of them. The travel poles are infinite in number and accumulate at $q = 1$, that is at $|x| = 1$ (Proposition 6 and Lemma 7); this is the boundary of the disc $|x| < 1$ of meromorphy, not of the disc of convergence $|x| < x_1$, which only the dominant pole bounds. They are locally finite in $(0, 1)$ (Lemma 7), so an infinite subset of them accumulates at $q = 1$ and nowhere else. Hence V has infinitely many poles in $|q| < 1$ accumulating at $|q| = 1$; under $q = x^2$ these are infinitely many poles in $|x| < 1$ accumulating at $|x| = 1$. An algebraic function over $\mathbb{Q}(x)$ has only finitely many poles in $|x| < 1$, so an infinite pole set accumulating at the boundary is impossible for one; hence V is transcendental. $\square$

Corollary 19. *The exponential growth rate of v_n is $\beta_2 = 1.4916177871\dots$, with β_2^2 the reciprocal of the smallest travel pole q_1 ; the value $\frac{3}{2}$ is excluded. It needs only the location of the dominant travel pole: not [5, Remark “the sharp size of T_2 ; HEURISTIC”, lem:cos], not [5, Appendix app:annulus], and not [5, Appendix app:star]. It does share with Theorem 18 the transfer model (M), which is what places q_1 on V rather than on G_0^T alone, and at the dominant pole the junction pairing is positive (Corollary 16). The lower bound β_2 for u_n and v_n is free even of (M), coming from the pure-travel subcount [3]. The same rate for u_n follows from Proposition 1 (the former hypothesis (T) of Theorem 20, now proved) together with the positivity of $\Pi_q(q_1)$; $u_n \leq v_n$ gives the upper bound unconditionally.*

Whether the number β_2 is transcendental is a separate question, open, and with a clean reduction to a value statement for a Hahn–Exton q -Bessel function; it is treated in [5, Appendix app:beta2], which is independent of everything else in this paper.

5.1 The true series U

Theorem 20. *The growth series U of W_{univ} is transcendental over $\mathbb{Q}(x)$. (Standing: Theorems 12 and 15.)*

Remark 21 (the inputs of Theorem 20). The travel poles pass from V to U by Proposition 1, whose proof does not use the metric identity. The factorisation [5, (eq:mobius)] with the dictionary [5, (eq:dict)] and the reciprocity $t_0 = b_1$ is [5, Theorem “the bulk dictionary at the travel poles”, thm:L]. The residue is the square of $\Pi_q(q_m)$, which is Theorem 15. The history of the former inputs (T) and (L) is recorded in [5, §§sec:inputT–sec:inputL].

Independently of (M), (T) and (R-J), the half $|s| < 1$ of the gate (7) holds at every travel pole. The numerator amplitude estimate $(\star)$ (at every travel pole q_m , $|P_{12}(q_m)| w_m^3 < 2$) is [5, Theorem “the amplitude estimate $(\star)$ ”, thm:star] ([5, Appendix app:star]): $|P_{12}| \leq 0.619 \tau^{3/2}$ for $\tau \leq 5 \cdot 10^{-3}$ and $|P_{12}| w^3 \leq 0.75$ at the six poles above that threshold. The passage to $|s| < 1$ is [5, Corollary “the gate closes”, app:gate]: $|s| \leq 0.983$ on the uniform range and $|s| \leq 0.296$ at the six exceptional poles, with the denominator lower bound $|S_e| \geq 0.63\sqrt{\tau}$ supplied by [5, Lemma “the denominator at the pole”, app:Se]. The remaining half $b_0 > 0$ is proved in the same corollary from a sign chain through the closed form for b_0 , which is [5, Corollary “the dictionary”, cor:dict]. The leading-order picture of [5, Remark “derivation of the U -numerator amplitude”, rem:elemY3] (the phase-matched zero giving $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$, a fourfold margin below the threshold $1/\sqrt{2}$) is consistent with, and explained by, the second-difference identity of [5, Proposition prop:P12second].

Proof. By Lemma 13 at $y = x^2$, U is meromorphic on $|x| < 1$, and the local forms below are those of this continuation. By Proposition 1 (the former input (T)), U inherits the travel poles. By Proposition 14 at $y = q$, at each travel pole U has the local form $\mathcal{N}_q(q - q_m)^{-\pi} + (\text{lower order}) + H_m$ with $\mathcal{N}_q = (2x_m)^{-1} c_\star \Pi_q(q_m)^2$ and $c_\star \neq 0$; the boundary vectors are holomorphic there because the gate (7) holds at *every* travel pole. Indeed, on the uniform range $\tau \leq 5 \cdot 10^{-3}$, $|P_{12}| \leq 0.619 \tau^{3/2}$ ([5, Theorem “the amplitude estimate $(\star)$ ”, thm:star]) and $|S_e| \geq 0.63\sqrt{\tau}$ ([5, Lemma “the denominator at the pole”, app:Se]) give $|s| = \frac{q}{1-q} |P_{12}|/|S_e| \leq \frac{\tau}{e^\tau - 1} \cdot \frac{0.619}{0.63} \leq 0.983 < 1$, and $b_0 > 0$ follows from the sign chain of the same lemmas through the closed form of [5, Corollary “the dictionary”, cor:dict]; the six exceptional poles are checked directly ([5, Corollary “the gate closes”, app:gate]). By Theorem 15 the leading coefficient $\mathcal{N}_q$ is non-zero at every q_m , so U has an uncanceled pole at each of them; these accumulate at $q = 1$ (Proposition 6 and Lemma 7), so U has infinitely many poles in the unit disc accumulating at its boundary and is transcendental over $\mathbb{Q}(x)$, as in Theorem 18. Transcendence does not consume the exceptional-pole enumeration of [5, Theorem “the amplitude estimate $(\star)$ ”, thm:star]: the uniform range $\tau \leq 5 \cdot 10^{-3}$ alone carries infinitely many poles (by Proposition 6 and Lemma 7, only finitely many travel poles lie outside it); the enumeration only upgrades “at infinitely many travel poles” to “at every travel pole” in the gate. For orientation: the computed values over $m = 5, \dots, 71$ are $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2}) = 0.1768$ and $s \rightarrow \frac{1}{4}$, well inside every bound above. $\square$

Remark 22 (the standing of Theorem 20). The unconditional content of this section is the gate: at every travel pole, $|P_{12}| \leq 0.619 \tau^{3/2}$ ([5, Theorem “the amplitude estimate $(\star)$ ”, thm:star]), $|S_e| \geq 0.63\sqrt{\tau}$ ([5, Lemma “the denominator at the pole”, app:Se]) and hence $|s| \leq 0.983 < 1$ ([5, Corollary “the gate closes”, app:gate]), together with the infinitude and accumulation of the travel poles (Proposition 6, Lemma 7) and the fact that these facts compose to non-algebraicity ([5, Remark “the assembly is machine-checked”, rem:uasembly]). The factorisation [5, (eq:mobius)], the reciprocity $t_0 = b_1$, the dictionary [5, (eq:dict)] and the half $b_0 > 0$ of the gate are [5, Theorem “the bulk dictionary at the travel poles”, thm:L]; the junction pairing is Theorem 15.

(T) is now proved. Proposition 1 follows from [5, Corollary cor:localzero]: a pure-travel element has no gap edge, so by [5, Theorem “no gap edge forces $c = 0$ ”, thm:nogap] some relaxed-optimal realisation is a single open walk, which bounds $\ell_T \leq \ell_R$; the reverse inequality is immediate. The metric identity $\ell_T = \ell_R + 2c$ is not used.

(R-J) is Theorem 15, and (M) is Theorem 12. The standing of Theorem 20 is therefore that of its inputs: the gate ([5, Appendix app:star], which contains verified interval-arithmetic enclosures); the metric identity and the faithfulness of the model (Lean); the assembly (M3) (a hand proof, [5, Appendix app:M3prime], with two independent adversarial reviews and exact series agreement to x^{26}); and, for the statement at every travel pole rather than infinitely many, the interval certificate at the first twelve poles. The assembly is not formalised.

6 Beyond algebraicity: D -finiteness, functional equations and the natural boundary

Theorems 18 and 20 exclude algebraic equations. The pole set that proves them excludes much more, because it is infinite and lies inside the disc $|x| < 1$ on which U and V are meromorphic (Lemma 13), well beyond their radius of convergence $x_1 = 0.6704\dots$. This section records what it excludes. Every statement below about U or V has the standing of Theorems 18 and 20 (Theorems 12 and 15); every statement about the travel resolvent G_0^T alone is unconditional, as Theorem 17 is.

6.1 Poles of U and V on the negative axis

Proposition 23. *At every travel pole q_m , U has a pole at $x = -x_m$ as well as at $x = x_m$, both of order π , the order of the pole of $(I - \mathcal{T})^{-1}$ at q_m . At $x = -x_m$ the junction pairing is negative:*

$$\Pi_q|_{x=-x_m} = \frac{2q_mB_0(1-x_m)}{1+x_m} \left(t_1 \frac{1+x_m+q_m}{1+q_m} - \frac{1}{2x_m} \right) < 0, \quad B_0 = \frac{1}{1-\mathfrak{g}_U t_1}.$$

(Standing: Theorems 12 and 15; at the six poles with $\tau_m > 5 \cdot 10^{-3}$, the enumeration standing of [5, Theorem “the amplitude estimate ($\star$)”, thm:star].)

Proof. The travel resolvent $(I - \mathcal{T})^{-1}$, the bulk vector Ψ , t_1 and B_0 depend on x only through $q = x^2$; x itself enters $\mathcal{W}(x, q)$ only through the monomials of [5, (eq:app-mu)–(eq:app-E)] and the prefactor x^{-1} . The proofs of Proposition 14 and of (11) use only $q = x^2$, never the sign of x (the reductions $x^{\max(2s+1, 2b+1)} = x q^{\max(\cdot, \cdot)}$ and the final polynomial identities hold for either root). Near $x = -x_m$ one has $q - q_m = (x + x_m)(x - x_m)$ with the second factor non-zero, so $U = \mathcal{N}_q(-2x_m)^{-\pi}(x + x_m)^{-\pi} + (\text{lower order})$ with $\mathcal{N}_q = c_\star(-2x_m)^{-1}\Pi_q|_{x=-x_m}^2$, and (11) at $x = -x_m$ is the displayed formula. The prefactor is positive: $0 < x_m < 1$, and $1 - \mathfrak{g}_U t_1 > 0$ because $|\mathfrak{g}_U t_1| < |\mathfrak{g}_V t_1| = |s| < 1$ ([5, Corollary “the gate closes”, app:gate]). For the bracket write $t_1 = s(1 - q)/q$ and use $(1 + x + q)/(1 + q) = 1 + x/(1 + x^2) \leq \frac{3}{2}$:

$$2x_m |t_1| \frac{1 + x_m + q_m}{1 + q_m} \leq 3|s| \frac{1 - q_m}{x_m}.$$

For $\tau_m \leq 5 \cdot 10^{-3}$, $|s| \leq 0.983$ and $(1 - q_m)/x_m \leq \tau_m e^{\tau_m/2} \leq 0.00502$, so the right side is below 0.015. At the six poles with $\tau_m > 5 \cdot 10^{-3}$, $|s| \leq 0.296$ ([5, Corollary “the gate closes”, app:gate]) and $(1 - q_m)/x_m \leq (1 - q_1)/x_1 = 0.8212$, since $q \mapsto (1 - q)/\sqrt{q}$ decreases; the right side is at most 0.730. In both cases $|t_1|(1 + x_m + q_m)/(1 + q_m) < 1/(2x_m)$, the bracket is negative, $\mathcal{N}_q \neq 0$, and $-x_m$ is a pole of order exactly π . $\square$

At the dominant pole the bracket is -0.2155 , and it tends to $-\frac{1}{2}$ along the ladder. Only the uniform range is needed for infinitely many poles at $-x_m$; the six-pole list upgrades this to every travel pole.

Proposition 24 (V at $-x_m$). *At every travel pole q_m , V has a pole at $x = -x_m$ of order π . With $\beta_m := t_1 - (1 - x_m)/(2x_m)$,*

$$(1 - \mathbf{g}_V t_1) \Pi_1|_{x=-x_m} = \frac{2q_m(1 - x_m)}{1 - q_m} \beta_m,$$

and for $\tau = \tau_m \leq 5 \cdot 10^{-3}$, with $h = \tau/2$, $\gamma = 4 \sinh h + 2 - 2 \cosh h$ and $\Lambda = e^{h/2} \sinh h / (2h(4 - \gamma/3))$,

$$\frac{1 - 5.14\sqrt{\tau}}{1 + 1.20\sqrt{\tau}} \leq \frac{\beta_m}{\Lambda\tau^2} \leq \frac{1 + 5.14\sqrt{\tau}}{1 - 1.20\sqrt{\tau}}, \quad \text{so} \quad 0.073\tau^2 \leq \beta_m \leq 0.187\tau^2.$$

At the six poles with $\tau_m > 5 \cdot 10^{-3}$, direct evaluation gives $\beta_m = 0.1168, 1.119 \cdot 10^{-3}, 1.362 \cdot 10^{-4}, 3.484 \cdot 10^{-5}, 1.266 \cdot 10^{-5}, 5.651 \cdot 10^{-6}$. (Standing: Theorems 12 and 15; at the six poles with $\tau_m > 5 \cdot 10^{-3}$, the enumeration standing of [5, Theorem “the amplitude estimate ($\star$)”, thm:star].)

Here the gate does not decide the sign, unlike for U : $t_1 = s(e^\tau - 1)$ with $s \rightarrow \frac{1}{4}$, and $(1 - x_m)/(2x_m) = (e^{\tau/2} - 1)/2 = \frac{\tau}{4} + O(\tau^2)$, so the two terms of β_m cancel at leading order. The proof computes the $O(\tau^2)$ remainder. It uses an exact second-order q -difference equation for the q -cosine to eliminate the unknown curvature at the zero.

Proof. The pole. $V = \mathcal{W}(x, 1)$, and the proof of Proposition 23 applies verbatim at $y = 1$: the local form of Proposition 14 and (10) use only $q = x^2$. Near $x = -x_m$ the leading coefficient is $c_\star(-2x_m)^{-1} \Pi_1|_{x=-x_m}^2$ with $c_\star \neq 0$, and (10) at $x = -x_m$ is the displayed identity. Its prefactor is positive, and $1 - \mathbf{g}_V t_1 = 1 - s > 0$ by the gate ([5, Corollary “the gate closes”, app:gate]). So $\Pi_1|_{x=-x_m} \neq 0$ as soon as $\beta_m \neq 0$, and then $-x_m$ is a pole of order exactly π . At the six exceptional poles, $\beta_m > 0$ is the listed evaluation. It remains to bound β_m for $\tau \leq 5 \cdot 10^{-3}$.

Notation. Let $c_j = (-1)^j q^{j^2+j} / (q; q)_{2j}$ and $C(y) = \sum_j c_j y^j$, and write $\psi(t) = \cos(Ze^t; q^2) = C(Z^2 e^{2t})$. Also write $\varphi(t) = e^{-t/2} \psi(t)$, $w = \sqrt{2/\tau}$, $A = \varphi'(0) = \psi'(0)$ and $M_k = \max_{|t| \leq \tau} |\varphi^{(k)}(t)|$. At the pole, $\psi(0) = 0$ (Proposition 2), so $\varphi(0) = 0$. Moreover $z_0 = Ze^{-h}$, $qZ = Ze^{-2h}$ and $x_m = e^{-h}$.

Step 1: β_m as a symmetric sum. We have $t_1 = P_{12}/S_e$ (§4.1), $S_e = \psi(-h)$ (Proposition 2) and $P_{12} = -\frac{1}{2}[\psi(h) + \psi(-h)]$ ([5, Proposition prop:P12second]). With $(1 - x_m)/(2x_m) = (e^h - 1)/2$ this gives

$$\beta_m = -\frac{\psi(h) + e^h \psi(-h)}{2\psi(-h)} = -\frac{e^{h/2} E_1}{2S_e}, \quad E_k := \varphi(kh) + \varphi(-kh). \quad (13)$$

Step 2: a q -difference equation. The coefficients satisfy $(1 - q^{2j-1})(1 - q^{2j})c_j = -q^{2j}c_{j-1}$. Summed against y^j , this gives $C(y) - (1 + q^{-1})C(q^2 y) + q^{-1}C(q^4 y) = -q^2 y C(q^2 y)$. In the variable t , $y \mapsto q^2 y$ is $t \mapsto t - 2h$. We shift by $2h$ and use $Z^2 = 2(e^{2h} - 1)$. Then we divide by $e^{t/2+h}$, which gives, for all real t ,

$$\varphi(t + 2h) + \varphi(t - 2h) = (2 \cosh h - 4 \sinh h e^{2t}) \varphi(t). \quad (14)$$

At $t = 0$, and after differentiating twice at $t = 0$, since $\varphi(0) = 0$:

$$E_2 = 0, \quad \varphi''(2h) + \varphi''(-2h) = (2 - \gamma) \varphi''(0) - 16 \sinh h A. \quad (15)$$

Step 3: elimination. Write $a_n = \varphi^{(n)}(0)/n!$. Taylor’s theorem with Lagrange remainders gives the following:

- $E_k = 2a_2(kh)^2 + 2a_4(kh)^4 + r_k$ with $|r_k| \leq 2(kh)^6 M_6/720$, for $k = 1, 2$;

- $\varphi''(2h) + \varphi''(-2h) = 4a_2 + 96a_4h^2 + r''$ with $|r''| \leq \frac{4}{3}h^4M_6$.

The two relations (15) determine a_2h^2 and a_4h^4 . Substituting them into E_1 gives, with $D = 4 - \gamma/3$, the exact identity

$$D E_1 = 4h^2 \sinh h A + \frac{1}{4}h^2 r'' - \left(\frac{\gamma}{16} + \frac{D}{4}\right)r_2 + D r_1.$$

Since $0 < \gamma < 12$, we have $0 < D < 4$, and the remainder is at most $C_6 h^6 M_6$ with $C_6 = \frac{\gamma}{90} + \frac{1}{3} + 4\left(\frac{2}{45} + \frac{1}{360}\right)$.

Step 4: the slope. By (15), $\varphi(2h) - \varphi(-2h) = -2\varphi(-2h) = -2e^h \cos(qZ; q^2)$. At a pole, $\cos(qZ; q^2) = qZ \sin(Z; q^2)$ and $|\sin(Z; q^2)| \geq \sqrt{1 - 0.61Z}$ for $Z \leq \min(0.2, q/3)$; both are proved in [5, proof of Proposition prop:selower], and here $Z \leq 0.1002$. Since $e^h qZ = z_0$, and Taylor's theorem gives $|\varphi(2h) - \varphi(-2h) - 4hA| \leq \frac{8}{3}h^3 M_3$, we get

$$|A| \geq \frac{z_0 |\sin(Z; q^2)|}{\tau} - \frac{\tau^2}{6} M_3 \geq w \left[\sqrt{\frac{1-q}{\tau}} \sqrt{1 - 0.61Z} - \frac{\tau}{3} \frac{M_3}{w^3} \right].$$

Step 5: derivative bounds. By [5, Lemma app:rep2] at radius $\rho = z_0 e^t$, $\psi(t) = \text{Re } \mathbb{E}[F]$. Here $F = (\rho e^{i\theta}; q)_{\infty}^{-1}$, $\theta = \frac{\pi}{2} + \sigma g$, $\sigma^2 = \tau/2$ and g is standard normal. We have $\partial_t F = -i\partial_{\theta} F$. Gaussian integration by parts, $\mathbb{E}[f^{(k)}(g)] = \mathbb{E}[f(g) \text{He}_k(g)]$ with He_k the Hermite polynomials, then gives

$$\psi^{(k)}(t) = \text{Re}[(-i)^k w^k \mathbb{E}[F \text{He}_k(g)]].$$

In $\log F = \sum_{n \geq 1} (\rho e^{i\theta})^n / (n(1 - q^n))$, the real parts are bounded term by term:

- $n = 1$: $-\rho \sin(\sigma g) / (1 - q) \leq \lambda |g|$, with $\lambda = \rho \sigma / (1 - q)$;
- $n = 2$: $-\kappa_2 \cos(2\sigma g) \leq -\kappa_2 + 2\kappa_2 \sigma^2 g^2$, with $\kappa_2 = \rho^2 / (2(1 - q^2))$, using $\cos 2u \geq 1 - 2u^2$;
- $n \geq 3$: at most $\rho^3 / (9q^2(1 - q)(1 - \rho)) \leq K_* := \rho^3 / (9q^2(1 - q)(1 - \rho/q))$, since $n(1 - q^n) \geq 3(1 - q^3) \geq 9q^2(1 - q)$.

Hence $|\psi^{(k)}(t)| \leq w^k e^{K_* - \kappa_2} \mathbb{E}[e^{\lambda|g| + \kappa_2 \tau g^2} |\text{He}_k(g)|]$. From $\varphi^{(k)} = e^{-t/2} \sum_j \binom{k}{j} (-\frac{1}{2})^{k-j} \psi^{(j)}$ and $1/(2w) = \sigma/2$, the same bound holds for $\varphi^{(k)}$ with the extra factor $e^{\tau/2}$, and with

$$\sum_j \binom{k}{j} (\sigma/2)^{k-j} |\text{He}_j(g)|$$

in place of $|\text{He}_k(g)|$. For $|t| \leq \tau$ and $\tau \leq \bar{\tau} = 5 \cdot 10^{-3}$, the parameters $\lambda, \rho, \sigma, \kappa_2 \tau, K_*$ are at most their values at $t = \tau = \bar{\tau}$. They are increasing in τ and t , and $\kappa_2 \geq e^{-2\bar{\tau}}/2$. Directed-rounding interval arithmetic evaluates the resulting Gaussian integrals on 6000 cells of $g \in [0, 14]$. The integrand is bounded beyond $g = 14$ by its value at 14, because it is decreasing there even after multiplication by e^g . The result is

$$\max_{|t| \leq \tau} |\psi''| \leq 3.24 w^2, \quad M_3 \leq 5.68 w^3, \quad M_6 \leq 53.3 w^6$$

(code/zeta_probe/nondfinite/vminusx_certificate.py, mpmath iv at 50 digits; the computed enclosures are 3.2196, 5.6320 and 52.193, rounded up to the stated constants, which are the ones used below).

Step 6: assembly. Each quantity below is monotone in τ , so it is evaluated at $\bar{\tau}$ with the rounded constants. Step 4 gives $|A| \geq 0.958 w$. By Taylor's theorem, $S_e = \psi(-h) = -hA(1 + \delta_2)$ with $|\delta_2| \leq hK_2 w / (2 \cdot 0.958) \leq 1.20\sqrt{\tau}$, where $K_2 = 3.24$. Step 3 and $\sinh h \geq h$ give $E_1 = (4h^2 \sinh h/D) A(1 + \delta_1)$ with $|\delta_1| \leq C_6 \cdot 53.3 \sqrt{\tau} / (4\sqrt{2} \cdot 0.958) \leq 5.14\sqrt{\tau}$. By (13),

$$\beta_m = \frac{e^{h/2} 4h^2 \sinh h}{2hD} \cdot \frac{1 + \delta_1}{1 + \delta_2} = \Lambda \tau^2 \frac{1 + \delta_1}{1 + \delta_2},$$

which is the two-sided bound. Finally, $\frac{1}{8} < \Lambda \leq \Lambda(\bar{\tau}) \leq 0.12527$, since $e^{h/2}$, $\sinh h/h$ and $1/D$ are increasing, and $\sqrt{\bar{\tau}} = 0.0707$. This gives $0.073\tau^2 \leq \beta_m \leq 0.187\tau^2$. In particular $\beta_m > 0$. $\square$

The six exceptional values and the uniform-range bracket are checked against direct evaluation at the travel poles in `code/zeta_probe/nondfinite/vminusx_check.py`, in floating point at 45 digits. Along the ladder, $\beta_m/\tau_m^2 \rightarrow \frac{1}{8}$: the value is 0.12560 at $m = 7$ and 0.12507 at $m = 19$, and $\beta_m/(\Lambda\tau_m^2) - 1$ is below $3 \cdot 10^{-3}$ there, against the certified 0.49 at $\bar{\tau}$. The error terms are lossy by a factor of about 10^2 ; the leading constant is not. The identity $\cos(qZ; q^2) = qZ \sin(Z; q^2)$ is checked there, and (14) in `vminusx_diff_eq_check.py`, both to working precision. Only the uniform range is needed for infinitely many poles of V at $-x_m$.

6.2 D -finiteness

Lemma 25. *Let $\Omega \subset \mathbb{C}$ be a domain, let $a_0, \dots, a_r, b$ be holomorphic on Ω with $a_r \neq 0$, and let f be meromorphic on Ω with $a_r f^{(r)} + \dots + a_1 f' + a_0 f = b$ off the poles of f . Then every pole of f in Ω is a zero of a_r .*

Proof. Let z_0 be a pole of f of order $k \geq 1$. Then $f^{(j)}$ has a pole of order exactly $k + j$ at z_0 , so for $j < r$ the term $a_j f^{(j)}$ has a pole of order at most $k + r - 1$, and b is holomorphic. If $a_r(z_0) \neq 0$, the term $a_r f^{(r)}$ has a pole of order exactly $k + r$, which nothing else in the equation cancels. Hence $a_r(z_0) = 0$. $\square$

Theorem 26. *U and V are not D -finite: neither satisfies a linear differential equation*

$$\sum_{j=0}^r a_j(x) f^{(j)}(x) = b(x)$$

with polynomial coefficients and $a_r \neq 0$, nor even one whose coefficients a_j and b are holomorphic on a neighbourhood of the closed disc $|x| \leq 1$. Equivalently, (u_n) and (v_n) are not P -recursive. The same holds for the travel resolvent G_0^T , as a series in q and as a series in x , and for G_0^T it is unconditional. (Standing for U and V : Theorems 12 and 15.)

Proof. Let f be one of the series and suppose it satisfies such an equation. By Lemma 13 (for G_0^T , [5, Theorem thm:blocks], first clause) f continues to a meromorphic function on the open unit disc D . The expression $\sum_j a_j f^{(j)} - b$ is then meromorphic on D and vanishes near 0, so by the identity theorem it vanishes on D off the poles of f . By Lemma 25 every pole of f in D is a zero of a_r . Since $a_r \neq 0$ is holomorphic on a neighbourhood of the compact closed disc, its zeros there are isolated and hence finite in number. But f has infinitely many poles in D : V at every x_m (Theorem 18) and every $-x_m$ (Proposition 24), U at every $\pm x_m$ (Theorem 20 and Proposition 23), and G_0^T at every q_m in the variable q and at every $\pm x_m$ in the variable x (Theorem 17). This is a contradiction. D -finiteness of a power series is equivalent to P -recursiveness of its coefficients [29, §6.4]. $\square$

The pole argument does not reach differential algebraicity: $1/\cos(\pi/(1-x))$ has poles at $x = 1 - 2/(2n+1)$, $n \geq 1$, infinitely many in $|x| < 1$ and accumulating at $x = 1$, and satisfies an algebraic differential equation. Whether U and V are D -algebraic is open; we expect not.

Corollary 27. *The bivariate series $\mathcal{W}(x, y) \in \mathbb{Z}[y][[x]] \subset \mathbb{Z}[[x, y]]$ is not D -finite in (x, y) . (Standing: Theorems 12 and 15.)*

Proof. D -finiteness is preserved under substitution of algebraic power series without constant term [19]. If $\mathcal{W}$ were D -finite, the substitution $y = x^2$ would make $\mathcal{W}(x, x^2) = U$ D -finite ([5, Corollary cor:M3prime-growth]), contrary to Theorem 26. $\square$

6.3 Linear q -difference and Mahler equations

Proposition 28. *Let f be U or V , as series in x , or G_0^T , as a series in q , and let $a_0, \dots, a_r, b$ be polynomials with $a_0 a_r \neq 0$.*

- (i) *If $Q \in \mathbb{C}^\times$ is not a root of unity, then $\sum_{i=0}^r a_i(x) f(Q^i x) \neq b(x)$.*
- (ii) *If $k \geq 2$ is an integer, then $\sum_{i=0}^r a_i(x) f(x^{k^i}) \neq b(x)$.*
- (iii) *If $f = G_0^T$, as a series in q , and A_0, A_1, b are polynomials with A_0, A_1 not both zero, then $A_0(q) f(q) + A_1(q) f(-q) \neq b(q)$.*

For G_0^T all three statements are unconditional ((ii) uses the window statement of [5, Lemma lem:infpoles], hence the verified bound of [5, Lemma lem:T2abs]); for U and V the standing is that of Theorems 18 and 20. The other cases with Q a root of unity are open, and for U and V this includes $Q = -1$: their pole sets are invariant under $x \mapsto -x$ (Propositions 23 and 24), and the argument below gives nothing. Remark 29 records what such a relation would force.

Proof. Let D be the open unit disc, P the pole set of f in D , and $\rho > 0$ the radius of convergence of f ; P lies in $\rho \leq |x| < 1$, has finitely many points in each disc $|x| \leq R < 1$, and is infinite (proof of Theorem 26). An equation as in (i) or (ii) holds as an identity of power series near 0.

(i) If $|Q| > 1$, replacing x by $Q^{-r}x$ and i by $r - i$ gives an equation of the same form with Q^{-1} in place of Q and new leading and trailing coefficients $a_r(Q^{-r}x)$, $a_0(Q^{-r}x)$; so we may assume $|Q| \leq 1$. Then every $f(Q^i x)$ is meromorphic on D , and the equation holds on D by the identity theorem. Let E be the finite zero set of a_0 in D . At $p \in P \setminus E$ the term $a_0 f$ is singular and b is not, so some $f(Q^i x)$ with $1 \leq i \leq r$ is singular at p , that is $Q^i p \in P$. Thus every $p \in P \setminus E$ has a successor $Q^i p \in P$, $1 \leq i \leq r$, and every point of P is the successor of at most r points. Follow successors from any $p \in P$; along the chain the exponent of Q increases strictly.

If $|Q| < 1$ the moduli decrease strictly, so the chain stays in the finite set $P \cap \{|x| \leq |p|\}$ and ends, necessarily at a point of E . A chain of n steps ends at modulus at most $|Q|^n |p| < |Q|^n$, and that modulus is at least ρ , so $n < N := \log \rho / \log |Q|$. Hence every point of P is reached backwards from E in fewer than N steps, and $|P| \leq |E| \sum_{n < N} r^n$ is finite.

If $|Q| = 1$ the chain stays on the circle $|x| = |p|$, which meets P in a finite set, and its points are pairwise distinct because Q is not a root of unity; so it ends in E . Hence P lies on the finitely many circles $|x| = |e|$, $e \in E$, each meeting P in finitely many points, and P is finite. Both cases contradict the infinitude of P .

(ii) Here every $f(x^{k^i})$ is meromorphic on D and the equation holds on D . Work on the positive axis: let $P_+ = P \cap (0, 1)$ and let E_+ be the finite zero set of a_0 in $(0, 1)$. As in (i), every $p \in P_+ \setminus E_+$ has a successor $p^{k^i} \in P_+$ with $1 \leq i \leq r$; it is smaller than p , so every chain ends in E_+ , and reading a chain backwards, $p = e^{k^{-n}}$ for some $e \in E_+$ and some integer $n \geq 0$. Hence, for $0 < R < 1$ and $\ell = \log(1/R)$,

$$\#(P_+ \cap (0, R]) \leq \sum_{e \in E_+} \#\{n \geq 0 : k^n \leq \log(1/e)/\ell\} \leq |E_+| \left(1 + \log_k \frac{L}{\ell}\right), \quad L = \max_{e \in E_+} \log(1/e),$$

which grows like $\log(1/\ell)$ as $R \uparrow 1$. On the other hand, by [5, Lemma “infinitude and accumulation of the travel poles”, lem:infpoles] there is a travel pole with $w \in (m\pi, (m+1)\pi)$ for every $m \geq 4$, at $x = e^{-1/w^2} < e^{-1/((m+1)\pi)^2}$, equivalently $q = e^{-2/w^2} < e^{-2/((m+1)\pi)^2}$, and it lies in P_+ (Theorems 18, 20, 17). Distinct windows give distinct poles, so $P_+ \cap (0, R]$ has at least $\lfloor c/(\pi\sqrt{\ell}) \rfloor - 4$ points, with $c = 1$ in the variable x and $c = \sqrt{2}$ in q . This grows like $\ell^{-1/2}$, faster than $\log(1/\ell)$, a contradiction for R close to 1.

(iii) Write $g = 1 - \Sigma_1$, so $f = \Sigma_0/g$ with Σ_0 and g holomorphic on $|q| < 1$. For $0 < p < 1$ every term of the series for g in the proof of Lemma 7 is positive at $q = -p$: $(-1)^j(-p)^{j^2} = p^{j^2}$, $1 - q = 1 + p$, and $(q; q)_{2j} = \prod_{i \leq 2j} (1 - (-p)^i)$ has the factors $1 + p^i$ (i odd) and $1 - p^i$ (i even). The term $j = 0$ is 1, so $g(-p) \geq 1$. Hence $f(-q) = \Sigma_0(-q)/g(-q)$ is holomorphic at every travel pole q_m . Suppose the relation holds. Both sides are meromorphic on the disc, so it holds there. At q_m the terms $A_1(q)f(-q)$ and b are holomorphic, and f has a pole (Theorem 17), so $A_0(q_m) = 0$. This holds at the infinitely many q_m , so $A_0 \equiv 0$. Then $A_1(q)f(-q) = b(q)$ with $A_1 \not\equiv 0$, which makes f rational, against its infinitely many poles. $\square$

In the variable x , $G_0^T(x^2)$ is even, and $G_0^T(x^2) - G_0^T((-x)^2) = 0$ is a relation of the form (i) with $Q = -1$. This is why (iii) is stated in q .

Remark 29 (roots of unity). Reduction. Let Q have exact order $N \geq 2$ and put $\zeta = Q$. A relation $\sum_i a_i(x)f(Q^i x) = b(x)$ is the same as $\sum_{j=0}^{N-1} A_j(x)f(\zeta^j x) = b(x)$ with $A_j = \sum_{i \equiv j \pmod{N}} a_i$. The hypothesis $a_0 a_r \not\equiv 0$ does not make this relation non-trivial: $a_0 = 1$, $a_N = -1$ gives $f(x) - f(Q^N x) = 0$ for every f . For a root of unity the question is therefore whether some relation with $A_0, \dots, A_{N-1}$ not all zero holds.

$Q = -1$ for U . Suppose $A_0(x)U(x) + A_1(x)U(-x) = b(x)$ with A_0, A_1 not both zero. By [5, Proposition prop:simplepoles], $\pi = 1$. By Proposition 14, with the factor x^{-1} of (8) and $q - q_m = (x - x_m)(x + x_m)$, U then has simple poles at $\pm x_m$ with residues $R_{\pm} = c_{\star} \Pi_q|_{x=\pm x_m}^2 / (4x_m^2)$. Both residues are non-zero (Theorem 15 and Proposition 23). The residues of $U(-x)$ at x_m and $-x_m$ are $-R_-$ and $-R_+$. Comparing residues gives, for every m ,

$$A_0(x_m)R_+ = A_1(x_m)R_-, \quad A_0(-x_m)R_- = A_1(-x_m)R_+.$$

If $A_1 \equiv 0$, the first identity gives $A_0(x_m) = 0$ for all m , so $A_0 \equiv 0$; likewise if $A_0 \equiv 0$. So neither A_i vanishes identically. Multiplying the two identities and dividing by $R_+ R_-$ gives $A_0(x)A_0(-x) = A_1(x)A_1(-x)$ at every x_m , hence identically. So $r = A_0/A_1$ satisfies $r(x)r(-x) = 1$ and, by (11) at $\pm x_m$,

$$r(x_m) = \rho_m := \frac{R_-}{R_+} = \left(\frac{1 - x_m}{1 + x_m} \right)^4 \left(\frac{b_-}{b_+} \right)^2, \quad b_{\pm} = t_1 \frac{1 \mp x_m + q_m}{1 + q_m} \pm \frac{1}{2x_m}, \quad (16)$$

at every m with $A_1(x_m) \neq 0$, that is, at all but finitely many m . The map $t_1 \mapsto b_-/b_+$ is a Möbius map with coefficients in $\mathbb{Q}(x_m)$ and determinant $1/x_m \neq 0$. So a relation forces $t_1(q_m) = \theta(x_m)$ for all large m , where θ is a branch of an algebraic function of degree at most 2 over $\mathbb{C}(x)$. For V the same argument gives $\rho_m^V = ((1 - x_m)/(1 + x_m))^2 (\beta_m/(t_1 + (1 + x_m)/(2x_m)))^2$, from (10) and Proposition 24, with the same conclusion. We have not excluded this. Leading asymptotics do not decide it: $16\rho_m/(1 - x_m)^4 \rightarrow 1$ and $16\rho_m^V/(1 - x_m)^6 \rightarrow 1$, and a finite jet of ρ_m at $x = 1$ is matched by a polynomial.

Numerics. If $\deg A_0, \deg A_1 \leq D$, (16) is a homogeneous linear system with one equation for each travel pole, $m = 1, \dots, 20$, and $2D + 2$ unknowns. For $D = 0, \dots, 9$ its smallest singular value, with rows and columns normalised, is non-zero. It falls from 0.88 at $D = 0$ to $6.6 \cdot 10^{-50}$ for U , and to $3.9 \cdot 10^{-50}$ for V , at $D = 9$. The values agree to 12 digits at 150- and 220-digit working precision. So no relation with $\deg A_i \leq 9$ holds (`code/zeta_probe/nondfinite/rootunity_fit.py`, floating point, not an interval certificate). The fast decay in D shows that ρ_m is very well approximated by rational functions of x_m . A test of this kind therefore cannot become a proof.

Other orders. Let $N \geq 3$. If U has no pole at $\zeta^j x_m$ for $\zeta^j \neq \pm 1$, only $U(x)$ and, for even N , $U(-x)$ are singular at x_m among the $U(\zeta^j x)$. For odd N this gives $A_0(x_m) = 0$. Applied at $\zeta^{-j} x_m$, it gives $A_j(\zeta^{-j} x_m) = 0$. If this holds for infinitely many m , then every A_j vanishes and there is no relation. For even N the same step leaves the pairs $(j, j + N/2)$, and each pair satisfies the

$Q = -1$ condition above with the polynomials $A_j(\zeta^{-j}x)$ and $A_{j+N/2}(\zeta^{-j}x)$. By Lemma 13, a pole at $\zeta^j x_m$ requires g , S_e or $1 - \mathbf{g}t_1$ to vanish at $q = \zeta^{2j} q_m$. For $\zeta^{2j} = -1$ the first cannot vanish, since $g(-p) \geq 1$ (proof of (iii)). The others are not controlled, and neither is g at the other roots of unity. At $m \leq 20$ all three are non-zero at ωq_m , for $\omega = -1$ and ω a primitive cube or fifth root of unity, and the same holds with $\mathbf{g}_V$ (`rootunity_check.py`, `rootunity_minus.py`). At $\omega = -1$ the evaluation cancels heavily: $S_e(-q_m) g(-q_m) \approx 1.171$, and $|S_e(-q_{20})| \approx 5 \cdot 10^{-715}$, computed at 1500 digits. The same caveat applies to G_0^T in q at roots of unity other than -1 .

The count in (ii) is made on the real axis on purpose. It uses only the travel poles that are proved to exist, and it needs no statement about where on the circle the poles of f accumulate; compare Remark 38. Mahler equations are those of [20]. The catalytic equation of §2 is a q -difference equation in the catalytic variable, not in x , and is not affected by (i).

6.4 The natural boundary

We use the following rationality theorem.

Theorem 30 (Pólya–Bertrandias). *Let $\Omega \subset \mathbb{C}$ be a simply connected domain containing 0 and stable under complex conjugation, and let φ be the Riemann map of the unit disc onto Ω with $\varphi(0) = 0$, $\varphi'(0) > 0$. If $\varphi'(0) > 1$ and $f \in \mathbb{Z}[[x]]$ is the Taylor series at 0 of a function meromorphic on Ω , then f is a rational function.*

This is Pólya’s theorem [25], which replaced the disc of Borel’s rationality theorem by a domain measured by transfinite diameter; Bertrandias [2] extended it to p -adic and adelic domains, whence the name. Bost and Chambert-Loir prove a generalisation to curves over number fields [7, Thm. 7.9], from which they recover the Pólya–Bertrandias criterion [7, Example 7.10], and Cantor’s rationality theorem [9, Thm. 5.2.2] is a capacity-theoretic generalisation. Meromorphy suffices, but the strict inequality $\varphi'(0) > 1$ is needed: for Ω the unit disc, $\varphi'(0) = 1$, and $\sum_n x^{2^n}$ is not rational.

Theorem 31. (a) *The travel resolvent $G_0^T = \Sigma_0/(1 - \Sigma_1) \in \mathbb{Z}[[q]]$ has $|q| = 1$ as a natural boundary. This is unconditional.*

(b) *U and V have $|x| = 1$ as a natural boundary. (Standing: Theorems 12 and 15.)*

More precisely, in each case there is no point ζ of the unit circle and no disc B about ζ such that the function continues meromorphically to $D \cup B$, D the open unit disc.

Proof. Let f be G_0^T (in q), U or V (in x). In each case f has integer coefficients (the travel-run counts, u_n, v_n), is meromorphic on D (Lemma 13), and is not rational, having infinitely many poles in D (Theorems 17, 18, 20). Suppose f continues meromorphically to $D \cup B$ with B the disc of radius $\delta > 0$ about ζ , $|\zeta| = 1$. Shrinking δ , we may assume $\delta < |\operatorname{Im} \zeta|$ when $\zeta \notin \mathbb{R}$, so that B and its conjugate $\bar{B}$ are disjoint; when $\zeta = \pm 1$, $\bar{B} = B$. Because the coefficients are real, $\overline{f(\bar{x})} = f(x)$ near 0, hence on D , and $x \mapsto \overline{f(\bar{x})}$ continues f meromorphically to $\bar{B}$, agreeing with f on $\bar{B} \cap D$. So f is meromorphic and single-valued on

$$\Omega = D \cup B \cup \bar{B},$$

which is stable under conjugation and simply connected: it is the disc D with one or two disjoint discs attached, each meeting D in a convex set. Let φ be its Riemann map, $\varphi(0) = 0$, $\varphi'(0) > 0$. By uniqueness of the Riemann map, $\overline{\varphi(\bar{z})} = \varphi(z)$, so φ has real Taylor coefficients (Schwarz reflection). The map $\psi = \varphi^{-1}|_D$ sends D into D with $\psi(0) = 0$ and is not onto, since $\Omega \neq D$; by the Schwarz lemma $0 < \psi'(0) < 1$, that is $\varphi'(0) > 1$. Theorem 30 now makes f rational, a contradiction. $\square$

A D -finite series continues analytically along every path avoiding the finitely many zeros of the leading coefficient of its equation, so Theorem 31 gives a second proof of the polynomial-coefficient case of Theorem 26. The first proof is elementary and also covers coefficients holomorphic near the closed disc.

The real travel poles accumulate only at $q = 1$. The next proposition gives a second accumulation point of poles of G_0^T on the circle, at $q = -1$, and Corollary 34 carries it over to U and V . Proposition 36 and Corollary 37 in §6.5 do the same at $q = \pm i$. None of these is needed for Theorem 31.

Proposition 32 (the zeros of $\cos(Z; q^2)$ accumulate at $q = -1$). *Write $q = -e^{-t}$ with $\operatorname{Re} t > 0$. Put $L = \log 4$, $\Lambda = -\pi^2 + i\pi L$, $\theta_1 = \arctan(\pi/L) = 66.19\dots^\circ$ and $t_j^0 = (L + i\pi)/(2j + \frac{3}{2})$, so that $\arg t_j^0 = \theta_1$. There is a j_0 such that for every $j \geq j_0$ the function $1 - \Sigma_1 = \cos(Z; q^2)$ has exactly one zero t_j^* in the disc $D_j = \{t : |1/t - 1/t_j^0| < 1/|\Lambda|\}$. This zero is simple, and*

$$t_j^* = t_j^0 + O(j^{-3}).$$

Hence $q_j^* = -e^{-t_j^*} \rightarrow -1$ along the ray $\arg t = \theta_1$. Their conjugates tend to -1 along $\arg t = -\theta_1$. Every q_j^* is a simple pole of $G_0^T = \Sigma_0/(1 - \Sigma_1)$. So the poles of G_0^T accumulate at $q = -1$, and in $x = q^{1/2}$ the poles of $G_0^T(x^2)$ accumulate at $x = \pm i$.

Proof. Write $1 - \Sigma_1 = \sum_{k \geq 0} b_k$ with $b_k = (-2(1 - q))^k q^{k^2} / (q; q)_{2k}$ (Proposition 2), and put $p = e^{-t}$, $t = re^{i\theta}$. Then $(q; q)_{2k} = (-p; p^2)_k (p^2; p^2)_k$ and $(-2(1 - q))^k q^{k^2} = (2(1 + p))^k p^{k^2}$.

Step 1: an exact decomposition. Let $L_t = \log(2(1 + p)) = L - \frac{t}{2} + O(t^2)$, and let

$$f(s) = e^{sL_t - ts^2} \frac{(-p^{2s+1}; p^2)_\infty (p^{2s+2}; p^2)_\infty}{(-p; p^2)_\infty (p^2; p^2)_\infty}.$$

This function is entire in s , and $f(k) = b_k$ for $k \geq 0$. Put $a = i\pi/t$. Since $p^{2a} = 1$,

$$f(s + a) = e^{-2\pi is} e^{(\pi^2 + i\pi L_t)/t} f(s). \quad (17)$$

Fix $K \geq 0$, and put $s_A = K + \frac{1}{2}$ and $x_A = s_A t$. We describe paths in the coordinate $x = st$. The path P_0 runs from x_A to $|x_A|$ along the circle $|x| = |x_A|$ and then along $[|x_A|, \infty)$. The segments C_1 , C_- and C_+ are $x_A + iy$ with $0 \leq y \leq \pi$, $x_A - iy$ with $0 \leq y \leq \pi$, and $x_A + iy$ with $0 \leq y \leq 2\pi$. Put $\gamma_- = C_- \cup (P_0 - i\pi)$ and $\gamma_+ = C_+ \cup (x_A + 2\pi i + e^{i\theta}[0, \infty))$. Since $\operatorname{Im} s = \operatorname{Im}(xe^{-i\theta})/r$, the path γ_- lies in $\operatorname{Im} s < 0$ and γ_+ in $\operatorname{Im} s > 0$, except at s_A . At the far ends of both paths f decays like $e^{-|x|^2 \cos \theta / r}$. The kernel $1/(1 - e^{-2\pi is})$ has residue $1/(2\pi i)$ at each integer. Integrate it against f around the region between γ_- and γ_+ , closed at infinity through half-integers. This gives $\sum_{k > K} b_k = \int_{\gamma_-} f/(1 - e^{-2\pi is}) ds + \int_{\gamma_+} f e^{2\pi is}/(1 - e^{2\pi is}) ds$. Expand each kernel once, as $1 + e^{-2\pi is}/(1 - e^{-2\pi is})$ and $e^{2\pi is} + e^{4\pi is}/(1 - e^{2\pi is})$. The integrands f and $f e^{2\pi is}$ are entire, so their integrals can be moved to P_0 and to $C_1 \cup (P_0 + i\pi)$. Then (17) on $P_0 + i\pi$ gives

$$1 - \Sigma_1 = H + J_0(1 + e^{\Lambda_t/t}) + \operatorname{conn}_1 + E_- + E_+, \quad \Lambda_t = -\pi^2 + i\pi L_t, \quad (18)$$

where $H = \sum_{k \leq K} b_k$, $J_0 = \int_{P_0} f ds$, $\operatorname{conn}_1 = \int_{C_1} f e^{2\pi is} ds$, $E_- = \int_{\gamma_-} f e^{-2\pi is} (1 - e^{-2\pi is})^{-1} ds$ and $E_+ = \int_{\gamma_+} f e^{4\pi is} (1 - e^{2\pi is})^{-1} ds$. On the shifted pieces, (17) gives $f(s - a)e^{-2\pi i(s-a)} = e^{(-\pi^2 - i\pi L_t)/t} f(s)$ and $f(s + 2a)e^{4\pi i(s+2a)} = e^{(-4\pi^2 + 2\pi i L_t)/t} f(s)$. The identity (18) is exact for every t with $\operatorname{Re} t > 0$ and every K . We also checked it numerically, to relative error 10^{-12} , for $K = 1, \dots, 6$.

Step 2: uniform approximation. Let $\Phi(x) = xL - x^2 - \frac{1}{4} \text{Li}_2(e^{-4x}) + \frac{\pi^2}{24}$. For fixed $\delta, \varepsilon > 0$, uniformly in $\text{Re } x \geq \delta$, $|\arg t| \leq \frac{\pi}{2} - \varepsilon$ and $|t| \leq 1$,

$$\log f(x/t) = \frac{\Phi(x)}{t} - \frac{x}{2} - \frac{1}{2} \log(1 - e^{-2x}) - \frac{1}{2} \log \frac{\pi}{t} + O(|t|(1 + |x|)). \quad (19)$$

Indeed, $\log(p^{2s+2}; p^2)_\infty = \sum_{m \geq 0} g(x + (m+1)t)$ with $g(y) = \log(1 - e^{-2y})$. Apply the Euler–Maclaurin formula in the real variable m . The error term is $O(|t| \int |g''(y)| |dy|)$, taken along the ray from $x+t$ in the direction θ . On that ray $\text{Re } y \geq \delta$, so the ray stays at distance at least δ from the singularities $i\pi\mathbb{Z}$ of g . The antiderivative is $-\frac{1}{2} \text{Li}_2(e^{-2y})$. The factor $(-p^{2s+1}; p^2)_\infty$ is treated in the same way with $\log(1 + e^{-2y})$, and the two dilogarithms combine by $\text{Li}_2(z) + \text{Li}_2(-z) = \frac{1}{2} \text{Li}_2(z^2)$. The normalisation uses the η -transformation $\log(p^2; p^2)_\infty = -\frac{\pi^2}{12t} + \frac{1}{2} \log \frac{\pi}{t} + O(|t|)$ and $\log(-p; p^2)_\infty = \frac{\pi^2}{24t} + O(|t|)$ (midpoint rule). Finally $x(L_t - L)/t = -\frac{x}{2} + O(|x||t|)$.

Step 3: the main term. On $[|x_A|, \infty)$ the function Φ is real and strictly concave. Its only critical point is where $\Phi'(x) = L - 2x - \log(1 - e^{-4x}) = 0$, that is $\sinh 2x = 2$. This is $x_0 = \frac{3}{2} \log \phi$, with $e^{-2x_0} = \sqrt{5} - 2$, ϕ the golden ratio. There $\Phi_0 = \Phi(x_0) = 0.8767334422\dots$ and $\Phi''(x_0) = -\sqrt{5}$. Laplace's method (Watson's lemma holds uniformly for a complex parameter $1/t$ with $|\arg t| \leq \frac{\pi}{2} - \varepsilon$ [23]) and (19) give $(\phi/5)^{1/4} e^{\Phi_0/t} (1 + O(t))$ for the integral over $[|x_A|, \infty)$. By Step 4 the arc of P_0 contributes $O(e^{(\Phi_0 \cos \theta - 0.14)/r})$. Hence

$$J_0 = (\phi/5)^{1/4} e^{\Phi_0/t} (1 + O(t)), \quad |J_0| \asymp e^{T/r}, \quad T = \Phi_0 \cos \theta.$$

Step 4: the landscape inequalities. Put $W_n(x) = \text{Re}((\Phi(x) + 2\pi i n x) e^{-i\theta})$, so that $|e^{(\Phi(x) + 2\pi i n x)/t}| = e^{W_n(x)/r}$. Take $|\theta - \theta_1| \leq 10^{-3}$ and $|x_A| \in [0.0495, 0.0505]$. Then:

- (a) $W_0 \leq T - 0.14$ on the arc of P_0 , and $W_1 \leq T - 0.10$ on C_1 , $W_{-1} \leq T - 0.10$ on C_- , $W_2 \leq T - 0.10$ on C_+ .
- (b) After the shifts, the exponent on $P_0 - i\pi$ is $W_0 - (\pi^2 \cos \theta + \pi L \sin \theta) \leq T - 7.9$. On the real axis this uses $W_0 = \Phi \cos \theta \leq T$. On the shifted ray the exponent is at most $L^2/(4 \cos \theta) + \pi^2/12 - 4\pi^2 \cos \theta + 2\pi L \sin \theta \leq T - 6.2$.
- (c) On $C_\pm$ the kernels satisfy $|1 - e^{\mp 2\pi i s}| = |1 + e^{-ve^{-i\theta}}|$ with $v = 2\pi y/r \geq 0$. This is at least $1 - e^{-\pi/(2 \tan \theta)}$, so the kernels are at most 2.01. On the shifted pieces they are at most 2 for small r .
- (d) For the head, $k \leq K$ gives $2k|t| \leq 2|x_A|$. Hence $|1 - p^{2i}| \geq 0.94 \cdot 2ir$, $\text{Re } p^{2i-1} > 0$ and $|2(1+p)| \leq 4$. So $|b_k| \leq 4^k/(k!(1.8r)^k) \leq e^{0.2417/r} \leq e^{(T-0.11)/r}$.

All of these are certified in interval arithmetic by `code/zeta_probe/rootunity_zeros/certify_landscape.py` (mpmath iv, 6.9s). The certificate subdivides each segment into cells. On each cell it uses the mean value theorem with the explicit bound $|\Phi'(x) + 2\pi i n| \leq L + 2|x| + \max(-\log(1 - \rho), \log 2) + \frac{\pi}{2} + 2\pi|n|$, where $\rho = e^{-4 \text{Re } x}$. It encloses Li_2 by its series with an explicit tail, and it carries θ and $|x_A|$ as intervals. On all these paths $\text{Re } x \geq 0.0199$, so (19) applies and

$$|H| + |\text{conn}_1| + |E_-| + |E_+| \leq C r^{-2} e^{(T-0.10)/r}.$$

Step 5: Rouché. Put $u = 1/t$ and $u_j = 1/t_j^0 = i\pi(2j + \frac{3}{2})/\Lambda$. For $t \in D_j$ we have $|t|/|t_j^0| = 1 + O(1/j)$ and $\arg t = \theta_1 + O(1/j)$. So if $K = K_j$ is chosen with $(K + \frac{1}{2})|t_j^0|$ nearest to 0.05, then for $j \geq j_0$ every $t \in D_j$ satisfies the hypotheses of Step 4 with this fixed K . Then J_0 is analytic and non-zero on D_j , and Steps 1–4 give $(1 - \Sigma_1)/J_0 = 1 + e^{\Lambda_t/t} + O(r^{-2}e^{-0.10/r})$. Since $L_t = L - \frac{t}{2} + O(t^2)$, we have $\Lambda_t/t = \Lambda u - \frac{i\pi}{2} + O(1/j)$ on D_j , so

$$\frac{1 - \Sigma_1}{J_0} = g_0(u) + O(1/j), \quad g_0(u) = 1 - ie^{\Lambda u} = 1 - e^{\Lambda(u - u_j)}.$$

For $|w| \leq 1$ we have $|1 - e^w| \geq |w|(1 - (e - 2)) > 0.28|w|$. So $|g_0| > 0.28$ on ∂D_j , where $|\Lambda(u - u_j)| = 1$. The zeros of g_0 are $2\pi/|\Lambda|$ apart, which exceeds the diameter $2/|\Lambda|$ of D_j , so u_j is the only zero of g_0 in D_j , and it is simple. By Rouché's theorem $1 - \Sigma_1$ has exactly one zero u^* in D_j , and it is simple. From $|g_0(u^*)| = O(1/j)$ we get $|u^* - u_j| = O(1/j)$, that is $t_j^* - t_j^0 = O(|t_j^0|^2/j) = O(j^{-3})$. Since $q = -e^{-t}$ is conformal, the zero is also simple in q .

Numerically, at 80 digits (`rootunity_zeros/zeros_rate.py`), $|t_j^* - t_j^0| j^3 = 0.087, 0.148, 0.178$ at $j = 3, 10, 60$. The evaluation of $\sum b_k$ at these points cancels heavily: the largest term is about $e^{c/|t|}$ times the sum, so the working precision must exceed this.

Step 6: the poles of G_0^T . Lemma 10(iii) is an identity between power series in u . Its coefficients are rational functions of q : the half-integer powers cancel, since $\mathfrak{s}(u) = \sum_j (-1)^j q^{j^2+2j} u^{2j+1} / (q; q)_{2j+1}$. The series converge for all u when $|q| < 1$. The identity holds for $0 < q < 1$, so it holds for all $0 < |q| < 1$. Lemma 9 is a termwise identity of power series in q that converge on $|q| < 1$, so it also holds there. The resulting pole invariant, and $\Sigma_0 S_0 = 2q/(1 - q)$, involve only $Z^2 = 2(1 - q)/q$ and $z_0^2 = 2(1 - q)$, so no branch has to be chosen. At $q = q_j^*$ this gives $\Sigma_0 S_0 = 2q_j^*/(1 - q_j^*) \neq 0$, hence $\Sigma_0(q_j^*) \neq 0$, and q_j^* is a simple pole of G_0^T . $\square$

The zeros q_j^* carry over to U and V once the bulk quantities are controlled there. The bulk denominator S_e and the numerator P_{12} are q -cosines of the same type as $1 - \Sigma_1$, and the resummation of Proposition 32 applies to them with one change: near $q = -1$ their terms alternate in sign relative to those of $1 - \Sigma_1$.

Proposition 33 (t_1 near $q = -1$). *Keep the notation of Proposition 32, and put*

$$S_{\pm}(q) = \sum_{j \geq 0} \frac{(-2(1 - q))^j q^{j^2 \pm j}}{(q; q)_{2j}}, \quad x_+ = x_0 + \frac{i\pi}{2}, \quad E_+ = \Phi_0 - \frac{\pi^2}{4} + \frac{i\pi L}{2},$$

so that $S_+ = S_e = \cos(z_0; q^2)$ and $S_- = \cos(z_0/q; q^2)$. There is $r_0 > 0$ such that, uniformly for $|\theta - \theta_1| \leq 10^{-3}$ and $0 < r \leq r_0$,

$$S_{\pm} = (5\phi)^{-1/4} e^{-i\pi/4} e^{\mp x_+} e^{E_+/t} (1 + O(t)). \quad (20)$$

In particular $S_e \neq 0$ there, and

$$t_1 = \frac{P_{12}}{S_e} = -\frac{1}{2} \left(1 + \frac{S_-}{S_e} \right) = \phi + O(t).$$

At the zeros of Proposition 32: $S_e(q_j^*) \neq 0$ for all large j , and $t_1(q_j^*) = \phi + O(1/j)$.

Proof. Step 0: t_1 as a ratio of two q -cosines. Write $c(u) = \cos(u; q^2)$, so that $S_e = c(z_0)$ (Proposition 2) and, from the series, $S_- = c(z_0/q)$. By [5, Lemma “the gate block in closed form”, lem:P12closed], $P_{12} = \frac{qZ}{2} \mathfrak{s}(Z) - c(z_0)$, and Lemma 10(i) at $u = z_0/q$ gives $c(z_0) - c(z_0/q) = qZ \mathfrak{s}(Z)$. Hence

$$P_{12} = -\frac{1}{2}(S_e + S_-)$$

on $(0, 1)$. This is the identity of [5, Proposition prop:P12second]; it is stated there at the travel poles, but these two steps do not use $c(Z) = 0$. Every term is holomorphic on $|q| < 1$: $qZ \mathfrak{s}(Z) = 2(1 - q) \sum_j (-1)^j q^{j^2+j} (2(1 - q))^j / (q; q)_{2j+1}$ involves only Z^2 . So the identity holds on $|q| < 1$. Since $t_1 = P_{12}/S_e$ as meromorphic functions ([5, Corollary cor:cocycle] and continuation), $t_1 = -\frac{1}{2}(1 + S_-/S_e)$ wherever $S_e \neq 0$.

Step 1: an exact decomposition. With p, b_k and f as in the proof of Proposition 32, the k -th term of $S_{\pm}$ is $b_k q^{\pm k} = (-1)^k p^{\pm k} b_k$. So $S_{\pm} = \sum_k (-1)^k F_{\pm}(k)$ with $F_{\pm}(s) = e^{\mp ts} f(s)$, which is

entire. We use the kernel $1/(2i \sin \pi s)$, whose residue at the integer k is $(-1)^k/(2\pi i)$. It equals $e^{-i\pi s}/(1 - e^{-2\pi i s})$ below the real axis and $-e^{i\pi s}/(1 - e^{2\pi i s})$ above it. With K , s_A , x_A and C_- as there (we take $r_0 \leq 10^{-3}$, so that some K has $|x_A| = (K + \frac{1}{2})r \in [0.0495, 0.0505]$, and fix such a K), and in the coordinate $x = st$, let C'_+ be $x_A + iy$ with $0 \leq y \leq \pi$. Put $\gamma'_\pm = C'_\pm \cup (x_A \pm i\pi + e^{i\theta}[0, \infty))$, where $C'_- = C_-$. These paths lie in $\pm \text{Im } s > 0$ except at s_A . The argument of Step 1 there gives

$$S_\pm = H_\pm + J_\pm^+ + J_\pm^- + E_\pm^+ + E_\pm^-.$$

Here $H_\pm = \sum_{k \leq K} (-1)^k F_\pm(k)$, $J_\pm^\nu = \int_{\gamma'_\nu} F_\pm e^{\nu i \pi s} ds$, and $E_\pm^\nu = \int_{\gamma'_\nu} F_\pm e^{3\nu i \pi s} (1 - e^{2\nu i \pi s})^{-1} ds$ for $\nu = \pm 1$. The integrands of $J_\pm^\pm$ are entire and decay in the sector between the directions 0 and θ of x . So we move $J_\pm^\pm$ to the path $V_\pm \cup \ell_\pm$. Here $V_\pm$ is the vertical segment from x_A to $\text{Re } x_A \pm \frac{i\pi}{2}$, and $\ell_\pm = \{\xi \pm \frac{i\pi}{2} : \xi \geq \text{Re } x_A\}$.

Step 2: the landscape. Put $\Omega_\nu(x) = \text{Re}((\Phi(x) + i\pi\nu x)e^{-i\theta})$ and $T' = \text{Re}(E_+ e^{-i\theta})$. At $\theta = \theta_1$, $T' = (\Phi_0 + \pi^2/4) \cos \theta_1 = 1.3500\dots$, which exceeds T by about 1. Since $\text{Li}_2(e^{-4x})$ has period $\frac{i\pi}{2}$ in x ,

$$\Phi(x \pm \frac{i\pi}{2}) \pm i\pi(x \pm \frac{i\pi}{2}) = \Phi(x) \pm \frac{i\pi L}{2} - \frac{\pi^2}{4}$$

exactly. On ℓ_+ this gives $\Omega_1 = \Phi(\xi) \cos \theta + T' - T \leq T'$. Equality holds only at $\xi = x_0$, which is the saddle x_+ . On ℓ_- it gives $\Omega_{-1} = \Phi(\xi) \cos \theta - \frac{\pi^2}{4} \cos \theta - \frac{\pi^2}{2} \sin \theta$. On the two rays, (17) turns the integrand of $E_\pm^\nu$ into $-e^{\nu i \pi s} e^{(-2\pi^2 + \nu i \pi L t)/t} F_\pm(s)$ with s real. There $\text{Re}(\Phi e^{-i\theta}) \leq L^2/(4 \cos \theta) + \pi^2/12$, as in Step 4(b). The following hold for $|\theta - \theta_1| \leq 10^{-3}$ and $|x_A| \in [0.0495, 0.0505]$:

$$\begin{aligned} \Omega_1 &\leq T' - 0.19 \text{ on } V_+, & \Omega_{-1} &\leq T' - 1.05 \text{ on } V_-, \\ \Omega_3 &\leq T' - 0.63 \text{ on } C'_+, & \Omega_{-3} &\leq T' - 0.79 \text{ on } C_-, \\ \Omega_{-1} &\leq T' - 3.9 \text{ on } \ell_-, & \text{ray exponents} &\leq T' - 3.2, \\ r \log |F_\pm(k)| &\leq T' - 1.1 \quad (k \leq K). \end{aligned}$$

The connector kernels are bounded by 2.01, as in Step 4(c), and the kernels on the rays by 2. All of these are certified in interval arithmetic by the script `certify_landscape_alt.py` in `code/zeta_probe/rootunity_zeros` (mpmath iv, 2.7 s). It uses the same cells, the same derivative bound with $2\pi|n|$ replaced by $\pi|\nu|$, and the same Li_2 enclosure as `certify_landscape.py`. On every path $\text{Re } x \geq 0.0199$, so (19) applies, and the factor $e^{\mp t s} = e^{\mp x}$ is bounded on the bounded pieces. On $\ell_\pm$ and on the rays it is at most $e^{|x|}$, which is dominated by the Gaussian decay of f ; on the head it adds at most $r|x_A|$ to $r \log |b_k|$. Hence

$$|H_\pm| + \left| \int_{V_+} \right| + |J_\pm^-| + |E_\pm^+| + |E_\pm^-| \leq C r^{-2} e^{(T' - 0.19)/r}.$$

Step 3: the main term. On ℓ_+ apply (19) and Laplace's method at $\xi = x_0$, as in Step 3, with $\Phi''(x_0) = -\sqrt{5}$. The factor $e^{\mp x}$ is exact and is evaluated at x_+ . So are $e^{-x_+/2} = \phi^{-3/4} e^{-i\pi/4}$ and $(1 - e^{-2x_+})^{-1/2} = (\sqrt{5} - 1)^{-1/2} = (\phi/2)^{1/2}$. This gives $\int_{\ell_+} F_\pm e^{i\pi s} ds = (5\phi)^{-1/4} e^{-i\pi/4} e^{\mp x_+} e^{E_+/t} (1 + O(t))$. Its modulus is $\asymp e^{T'/r}$. With Step 2 this is (20).

Step 4: the limit. By (20), $S_-/S_e = e^{2x_+}(1 + O(t))$, and $e^{2x_+} = -e^{2x_0} = -\phi^3$. So $t_1 = \frac{1}{2}(\phi^3 - 1) + O(t) = \phi + O(t)$, since $\phi^3 = 2\phi + 1$. At q_j^* , Proposition 32 gives $|t_j^*| \asymp 1/j$ and $\arg t_j^* - \theta_1 = O(j^{-2})$. With $K = K_j$ chosen as in its Step 5, the hypotheses above hold for all large j . $\square$

Numerically (`rootunity_zeros/t1_minusone.py`, 40–100 digits, floating point): the two expressions for t_1 agree to working precision at every zero tested. The ratio of S_e to the right side of (20)

is $0.9897 + 0.0331i$, $0.9933 + 0.0192i$ and $0.9962 + 0.0103i$ at $j = 10, 20, 40$. Also $(t_1 - \phi)/t_j^* = -0.1440 + 0.0387i$, $-0.1529 + 0.0197i$ and $-0.1574 + 0.0099i$ at $j = 40, 80, 160$. The next order of Laplace's method, computed formally, gives $t_1 = \phi - \frac{\phi}{10}t + O(t^2)$, and $\phi/10 = 0.1618$. This is consistent with the numerics but is not proved.

Corollary 34 (the poles of U and V accumulate at $x = \pm i$). *For all sufficiently large j , U and V have poles at $x = \pm\sqrt{q_j^*}$ and at $x = \pm\sqrt{\bar{q}_j^*}$. These tend to $x = \pm i$. At $q = q_j^*$,*

$$1 - \mathbf{g}_U t_1 = \frac{\phi}{2t_j^*}(1 + O(1/j)), \quad 1 - \mathbf{g}_V t_1 = 1 + \frac{\phi}{2} + O(1/j),$$

and, at the root $x \rightarrow \pm i$,

$$\Pi_q = -4 + O(1/j), \quad \Pi_1 = -\frac{2\phi(\phi \pm i)}{2 + \phi} + O(1/j).$$

(Standing: Theorems 12 and 15.)

Proof. Proposition 14 at a non-real zero. Let q^* be a zero of $1 - \Sigma_1$ with $0 < |q^*| < 1$. Suppose $S_e(q^*) \neq 0$ and $1 - \mathbf{g}(q^*, y)t_1(q^*) \neq 0$. Then the proof of Proposition 14 applies at q^* without change. The resolvent is meromorphic on $|q| < 1$ (Lemma 13). The kernel is at most one-dimensional and is normalised by $\sum_s q^s R_s = \Sigma_1(q^*) = 1$; this is [5, Lemma lem:nullspace], which is stated for $0 < |q| < 1$. The boundary vectors and $\mathcal{W}_0$ are holomorphic at q^* under the two conditions above (Lemma 13). And $\Sigma_0(q^*) \neq 0$ by Step 6 of the proof of Proposition 32. So $\Pi_y \neq 0$ at a root x^* of $x^2 = q^*$ makes x^* a pole of $\mathcal{W}(\cdot, y)$.

The closed forms at a non-real zero. The proof of (10)–(11) uses $\mathcal{T}R = R$ with $R \in \ell^1$, the normalisation $\sum_s q^s R_s = 1$, and $\Psi = R_0 u \in \ell^1$, which needs $S_e \neq 0$. It also uses absolute convergence of the double sums and polynomial identities in x with $x^2 = q$. It never uses $q > 0$; only the positivity paragraphs do. So both closed forms hold at q^* , for either root x . This is a re-reading of the proof, not an analytic continuation: R depends on the point. We checked it directly at $q_3^* = -0.7604 + 0.3406i$ and $q_4^* = -0.8181 + 0.2819i$ with truncated operators ($N = 186$ and 230). There $\langle \lambda^{(y)}, R \rangle$ agrees with the closed forms to relative error at most $5 \cdot 10^{-35}$ for $y \in \{1, q\}$ and both roots (`rootunity_zeros/pairing_complex_check.py`).

The asymptotics. Let $q = q_j^* = -e^{-t}$ with $t = t_j^*$. Then $1 + q = t(1 + O(t))$ and $1 - q = 2 + O(t)$. By Proposition 33, $S_e(q) \neq 0$ and $t_1 = \phi + O(t)$. Hence $\mathbf{g}_U t_1 = \frac{qt_1}{(1-q)(1+q)} = -\frac{\phi}{2t}(1 + O(t))$ and $\mathbf{g}_V t_1 = -\frac{\phi}{2} + O(t)$, which gives the two displayed denominators. Both are non-zero for large j . So $B_0 = \frac{2t}{\phi}(1 + O(t))$ at $y = q$. Then $B_0 t_1 \frac{1-x+q}{1+q} = 2(1-x+q)(1+O(t)) = -2x + O(t)$ and $B_0/(2x) = O(t)$, so (11) gives $\Pi_q = -4qx \frac{1+x}{1-x} + O(t)$. At $q = -1$ and $x = \pm i$ this is -4 . In (10), $\frac{2q(1+x)}{1-q} \rightarrow -(1+x)$, and $t_1 + \frac{1+x}{2x} \rightarrow \phi + \frac{1}{2} \mp \frac{i}{2}$; dividing by $1 + \frac{\phi}{2}$ gives the value of Π_1 .

Conclusion. For large j the hypotheses of the first paragraph hold at q_j^* for $y = 1$ and $y = q$, and $\Pi_1, \Pi_q \neq 0$ at both roots. So $\pm\sqrt{q_j^*}$ are poles of V and of U . The coefficients of U and V are real, so $\overline{f(\bar{x})} = f(x)$ on $|x| < 1$, and the conjugate points are poles as well. Finally, $q_j^* \rightarrow -1$ gives $\sqrt{q_j^*} \rightarrow \pm i$. $\square$

For U the exact value of t_1 does not matter. The limit -4 needs only $t_1 \asymp 1$, since $\mathbf{g}_U$ has a pole at $q = -1$ and t_1 cancels in Π_q . For V it does matter: Π_1 vanishes only if $t_1 = -(1+x)/(2x)$, and $1 - \mathbf{g}_V t_1$ vanishes only if $t_1 = -2$ in the limit. The limit ϕ is neither. Numerically all four pairings and both denominators are non-zero at every $j \geq 3$ tested, and approach the limits above at rate $O(1/j)$ (`t1_minusone.py`). The computer-assisted inputs are the two landscape certificates.

6.5 The point $q = i$

The resummation also works at $q = i$, with two changes. First, q^{k^2} depends on $k \bmod 2$, so the even and odd terms get separate interpolants. Second, two saddles of equal height compete, so the zeros come from a cancellation between two saddle contributions and not from the quasi-periodicity (17). Throughout, $q = ip$, $p = e^{-t}$, $P = p^4$ and $t = re^{i\theta}$ with

$$|\theta - \theta_i| \leq 10^{-3}, \quad \theta_i = \arctan \frac{\pi}{6 \log 2} = 37.067 \dots^\circ, \quad 0 < r \leq 0.005.$$

Put $a = -2(1 - q)$, $c_0 = -2 + 2i$, $L_w = \log c_0 = \frac{3}{2} \log 2 + \frac{3\pi i}{4}$ and $\ell(t) = \log(1 - \frac{1-i}{2}(1 - e^{-t}))$, so that $\log a = L_w + \ell(t)$. The functions

$$f_e(s) = \frac{e^{2s \log a - 4ts^2}}{(q; q)_\infty} \prod_{j=1}^4 (i^j p^{4s+j}; P)_\infty, \quad f_o(s) = \frac{i e^{(2s+1) \log a - t(2s+1)^2}}{(q; q)_\infty} \prod_{j=3}^6 (i^j p^{4s+j}; P)_\infty$$

are entire, $f_e(k) = b_{2k}$ and $f_o(k) = b_{2k+1}$ for $k \geq 0$, and both vanish at the negative integers. We use the coordinate $x = 2st$ for f_e and $x = (2s+1)t$ for f_o , so that $x = kt$ at the k -th term, and write $f(x)$ for either one. Let

$$\Phi_i(x) = xL_w - x^2 - \frac{1}{16} \text{Li}_2(e^{-8x}) + \frac{\pi^2}{96},$$

$$h_\pm(x) = -\frac{(1-i)x}{2} + \frac{1}{4} \log(1 \mp ie^{-2x}) - \frac{1}{4} \log(1 \pm ie^{-2x}) - \frac{1}{2} \log(1 \mp e^{-2x}),$$

with principal logarithms, and for $\text{Re } x \leq 0$ let

$$\Upsilon(x) = \text{Re}\left((xL_w - x^2 + \frac{17\pi^2}{96} + \frac{1}{16} \text{Li}_2(e^{8x}))e^{-i\theta}\right) + \sum_{\varrho=0}^3 \max_{m \equiv \varrho \pmod{4}} \frac{1}{2} \text{Re}((x - \frac{i\pi m}{4})^2 e^{-i\theta}).$$

For $a' \geq 0$ and $\varphi \in [0, \pi]$ put $\mathcal{D}(a', \varphi) = \inf_{\rho \geq 0} \max\{\sigma(\varphi - \rho \sin \theta), 1 - e^{-a' - \rho \cos \theta}\}$, with $\sigma(u) = \sin \min(\text{dist}(u, 2\pi\mathbb{Z}), \frac{\pi}{2})$. Let

$$d(x) = \min_{c^4=1} \mathcal{D}(2 \text{Re } x, \text{dist}(\arg c - 2 \text{Im } x, 2\pi\mathbb{Z})),$$

$$d'(x) = \min_{c^4=1} \mathcal{D}(-2 \text{Re } x, \text{dist}(2 \text{Im } x - \arg c, 2\pi\mathbb{Z})).$$

Finally $K_n = \frac{1}{D_n} + \frac{1}{D_n^2 \cos \theta} + \frac{4.75}{D_n} + \frac{1}{6} + 10^{-3}$ with $D_n = \mathcal{D}(0, \frac{\pi}{2})$, and $c_\ell = \bar{r}^{-2}(\frac{\sqrt{2}}{2}(e^{\bar{r}} - 1 - \bar{r}) + \frac{\mu^2}{2(1-\mu)})$ with $\bar{r} = 0.005$ and $\mu = \frac{\sqrt{2}}{2}(e^{\bar{r}} - 1)$. Numerically $c_\ell \leq 0.607$, $D_n \geq 0.668$ and $K_n \leq 11.58$.

Lemma 35 (bounds for the interpolants near $q = i$). *In this setting the following hold for f_e and for f_o .*

(i) *If $\text{Re } x \geq 0$ and $d(x) > 0$, then, modulo $2\pi i$,*

$$\log f(x) = \frac{\Phi_i(x)}{t} + h_\pm(x) + i\pi(\frac{1}{8} + \frac{\epsilon}{2}) - \frac{1}{2} \log \frac{\pi}{2t} + E,$$

$$|E| \leq \left(c_\ell |x| + \frac{121}{12 d(x)} + \frac{4}{3 \cos \theta d(x)^2} + K_n\right) |t|,$$

with $+$, $\epsilon = 0$ for f_e and $-$, $\epsilon = 1$ for f_o .

(ii) *If $\text{Re } x \leq 0$ and $d'(x) > 0$, then $\log |f(x)| \leq \Upsilon(x)/r + M'(x) + \frac{1}{2} \log \frac{2r}{\pi} + 4 \log(3 + 3e^{-\pi^2 \cos \theta/r}) + K'(x)r$, where $M'(x) = (\frac{1}{\sqrt{2}} + 2)|x| + \frac{\pi}{2}(|m^*| + 4) + 2\pi + \max(-\log d'(x), \log 2)$, $m^* = 4 \text{Im}(xe^{-i\theta})/(\pi \cos \theta)$ and $K'(x) = c_\ell |x| + \frac{73}{12 d'(x)} + \frac{4}{3 \cos \theta d'(x)^2} + \frac{3}{4} + K_n$.*

(iii) If $\operatorname{Re} x \geq X > 0$, then $r \log |f(x)| \leq \operatorname{Re}((xL_w - x^2 + \frac{\pi^2}{96})e^{-i\theta}) + \frac{e^{-2X}}{(1-e^{-2X})\cos\theta(1-2r\cos\theta)} + r(\frac{|x|}{\sqrt{2}} + c_\ell r|x| + \frac{1}{2} \log \frac{2r}{\pi} + K_n r)$.

(iv) Let $\Gamma_\pm$ be polygonal paths leaving a half-integer, whose pieces have directions $\arg(dx e^{-i\theta})$ in a closed cone inside $(0, \pi)$ for Γ_+ and inside $(-\pi, 0)$ for Γ_- , and let $\alpha_{\min}$ be the least angle between the cone and the real axis. Then $|1 - e^{\pm 2\pi i s}|^{-1} \leq (1 - e^{-(\pi/2)\tan\alpha_{\min}})^{-1}$ on $\Gamma_\pm$.

Proof. A ray lemma. Let $|c| = 1$, $G(v) = \log(1 - ce^{-v})$, $|ce^{-v_0}| \leq 1$, and $\delta = \inf_{\rho \geq 0} |1 - ce^{-v_0 - \rho e^{i\theta}}| > 0$. Euler–Maclaurin of order two for $m \mapsto G(v_0 + 4tm)$ along the ray $v_0 + e^{i\theta}\mathbb{R}_{\geq 0}$ gives $\sum_{m \geq 0} G(v_0 + 4tm) = -\operatorname{Li}_2(ce^{-v_0})/(4t) + \frac{1}{2}G(v_0) + E_0$ with $|E_0| \leq \frac{|t|}{3}(\frac{1}{\delta} + \frac{1}{\delta^2 \cos\theta})$, because $|G'| \leq 1/\delta$ and $\int |G''| \leq 1/(\delta^2 \cos\theta)$ on the ray. Moving the base point by h costs $hG(v) + O(\frac{|h|^2}{2\delta})$, since $\frac{d}{dv} \operatorname{Li}_2(ce^{-v}) = G(v)$. Along the ray $w = ce^{-v}$ has $|w| \leq e^{-a' - \rho \cos\theta}$ and $|1 - w| \geq \max(1 - |w|, \sigma(\arg w))$, and the argument of w turns at rate $\sin\theta$; this gives $\delta \geq \mathcal{D}(a', \varphi)$ with $e^{-a'} = |w_0|$ and $\varphi = \operatorname{dist}(\arg w_0, 2\pi\mathbb{Z})$, in either direction of turning.

Part (i). For f_e , $\log(i^j e^{-2x-jt}; P)_\infty = \sum_m G_j(2x + jt + 4tm)$ with $c = i^j$, and $\delta \geq d(x)$. The ray lemma, the shift from $2x + jt$ to $2x$, and $\sum_{j=1}^4 \operatorname{Li}_2(i^j z) = \frac{1}{4} \operatorname{Li}_2(z^4)$ give $-\operatorname{Li}_2(e^{-8x})/(16t)$ and the term $\sum_j (\frac{1}{2} - \frac{j}{4})G_j(2x)$, which is the logarithmic part of h_+ . The normalisation $(q; q)_\infty = \prod_{j=1}^4 (i^j e^{-jt}; P)_\infty$ is treated by the ray lemma from $v = 0$ for $j \leq 3$ ($\delta \geq D_n$), which contributes $-\frac{\pi^2}{32t} + \frac{i\pi}{8}$ to $-\log(q; q)_\infty$, and by the η -transformation for $j = 4$, $\log(P; P)_\infty = -\frac{\pi^2}{24t} + \frac{1}{2} \log \frac{\pi}{2t} + O(|t|)$. Finally $x \log a/t = xL_w/t - \frac{(1-i)x}{2} + O(c_\ell |x||t|)$. The branch f_o is the same with $c = -i^{j-2}$, which gives h_- , and its factor i gives $\frac{i\pi}{2}$. The error constants add up to the stated bound.

Part (ii). Write $(z; P)_\infty = \vartheta(z)/(P/z; P)_\infty$ with $\vartheta(z) = (z; P)_\infty (P/z; P)_\infty$. The triple product and Poisson summation give $(P; P)_\infty \vartheta(e^{-v}) = \sqrt{\pi/(2t)} \sum_{k \in \mathbb{Z}} \exp(\beta_k^2/(8t))$ with $\beta_k = 2t - v + i\pi(1 - 2k)$. With $v = 2x + jt - \frac{i\pi\gamma_j}{2}$ ($c_j = e^{i\pi\gamma_j/2}$) the exponent is $(x - \frac{i\pi m}{4})^2/(2t) - \frac{2-j}{2}x - \frac{i\pi m}{4} + \frac{(2-j)^2 t}{8}$, $m = \gamma_j + 2 - 4k$, and m runs over one class mod 4 for each j . The exponent is a concave quadratic in k with leading coefficient $-\pi^2 \cos\theta/(2r)$, so the sum is at most $3 + 3e^{-\pi^2 \cos\theta/r}$ times its largest term, which is one of the two class members next to m^* . The factors $(P/z_j; P)_\infty$ have argument $c_j^{-1} e^{2x} e^{-(4-j)t}$ of modulus ≤ 1 and are treated by the ray lemma with $\delta \geq d'(x)$. The $1/t$ terms add up to $\Upsilon(x)/r$, and the rest to M' and K' .

Part (iii) uses $-\log|1 - w| \leq |w|/(1 - |w|)$ termwise, with $|z_j| \leq e^{-2X}$ and $1 - |P| \geq 4r \cos\theta(1 - 2r \cos\theta)$. *Part (iv):* $s - s_c$ lies in the cone, and $e^{2\pi i s_c} = -1$ gives $|1 - e^{2\pi i s}| = |1 + e^{-A + iB}|$ with $A = |B| \tan\alpha' \geq 0$; this is ≥ 1 if $|B| \leq \frac{\pi}{2}$ and $\geq 1 - e^{-(\pi/2)\tan\alpha_{\min}}$ otherwise. $\square$

The Poisson form of the triple product was also checked to 10^{-28} at a sample point, and the error in (i) is $(0.13\text{--}0.58)|t|$ at the points tested, well below the bound. The full proof, with every constant, is in `rootunity_zeros/qi_bounds.tex`.

Proposition 36 (the zeros of $\cos(Z; q^2)$ accumulate at $q = i$). Write $q = ie^{-t}$ with $\operatorname{Re} t > 0$. Put

$$\Delta V = \frac{\pi^2}{8} - \frac{3\pi i}{4} \log 2, \quad t_j^0 = \frac{6 \log 2 + i\pi}{16j + 6 - 2i}, \quad u_j = 1/t_j^0,$$

so that $\arg t_j^0 = \theta_i + O(1/j)$. There is a j_0 such that for every $j \geq j_0$ the function $1 - \Sigma_1 = \cos(Z; q^2)$ has exactly one zero t_j^* in the disc $D_j = \{t : |1/t - u_j| < 1/|\Delta V|\}$. This zero is simple, and

$$t_j^* = t_j^0 + O(j^{-3}).$$

Hence $q_j^* = ie^{-t_j^*} \rightarrow i$ along the ray $\arg t = \theta_i$, and the conjugates $\bar{q}_j^* \rightarrow -i$. Every q_j^* and every $\bar{q}_j^*$ is a simple pole of $G_0^T = \Sigma_0/(1 - \Sigma_1)$. In $x = q^{1/2}$ the poles of $G_0^T(x^2)$ accumulate at the four

points $\pm e^{\pm i\pi/4}$. More precisely, with $x_0 = \frac{1}{4} \log(4 + \sqrt{15}) + \frac{3\pi i}{8}$ and $x_- = x_0 - \frac{i\pi}{2}$, uniformly for $|\theta - \theta_i| \leq 10^{-3}$ as $r \rightarrow 0$,

$$1 - \Sigma_1 = A_0 e^{V_0/t} (1 + O(t)) + A_- e^{V_-/t} (1 + O(t)) + O(e^{(T_i - 0.05)/r}), \quad (21)$$

where $V_0 = \Phi_i(x_0)$, $V_- = \Phi_i(x_-) - i\pi x_-$, $V_- - V_0 = \Delta V$, $T_i = \min(\operatorname{Re}(V_0 e^{-i\theta}), \operatorname{Re}(V_- e^{-i\theta}))$, and

$$A_0 = \left(\frac{2}{\sqrt{15}}\right)^{1/2} e^{i\pi/8} e^{h_+(x_0)} (1 + i\phi), \quad A_- = \left(\frac{2}{\sqrt{15}}\right)^{1/2} e^{i\pi/8} e^{h_+(x_-)} \left(1 - \frac{i}{\phi}\right), \quad \frac{A_-}{A_0} = e^{(1-i)\pi/4}.$$

Both amplitudes are non-zero.

Proof. Step 0: the saddles. Here $\Phi'_i(x) = L_w - 2x - \frac{1}{2} \log(1 - e^{-8x})$. With $a_0 = \frac{\sqrt{5}-\sqrt{3}}{2}$ we have $e^{-2x_0} = -a_0(1+i)$, $e^{-4x_0} = i(4 - \sqrt{15})$ and $e^{-8x_0} = -(4 - \sqrt{15})^2$, so $1 - e^{-8x_0} = 8(4 - \sqrt{15})$ and $\Phi'_i(x_0) = 0$ exactly. Since $\operatorname{Li}_2(e^{-8x})$ has period $\frac{i\pi}{4}$ in x ,

$$\Phi_i\left(x - \frac{i\pi}{2}\right) = \Phi_i(x) - \frac{i\pi L_w}{2} + i\pi x + \frac{\pi^2}{4},$$

so $\Phi'_i(x_-) = i\pi$ and $V_- - V_0 = -\frac{i\pi L_w}{2} - \frac{\pi^2}{4} = \Delta V$. Hence $\operatorname{Re}(\Delta V e^{-i\theta}) = \frac{\pi^2}{8} \cos \theta - \frac{3\pi}{4} \log 2 \sin \theta$, which vanishes exactly at $\theta = \theta_i$: there the two saddles have the same height, $-0.0706856\dots$. Also $\Phi''_i = -2 - 4e^{-8x}/(1 - e^{-8x}) = -\frac{\sqrt{15}}{2}$ at x_0 and at x_- .

Step 1: an exact decomposition. Fix a branch $F \in \{f_e, f_o\}$ and a half-integer $s_c < 0$ with x_c (the point s_c in the coordinate x) within 0.02 of $-\frac{\pi}{8}(\cot \theta_i + i)$. Since F vanishes at the negative integers, $\sum_{k \geq 0} F(k) = \int_{\Gamma_-} F K_- ds + \int_{\Gamma_+} F K_+ ds$ with $K_- = (1 - e^{-2\pi is})^{-1}$ and $K_+ = e^{2\pi is}(1 - e^{2\pi is})^{-1}$, where $\Gamma_{\mp}$ leave s_c in $\operatorname{Im} s \leq 0$. In the coordinate x : Γ_- runs from x_c to the line $\operatorname{Im} x = -\frac{\pi}{8}$ and along it to $+\infty$ through x_- ; Γ_+ runs from x_c to $\frac{i\pi}{8}$ and then along the ray $\frac{i\pi}{8} + e^{i \cdot 55^\circ} \mathbb{R}_{\geq 0}$. Expand $K_- = 1 + e^{-2\pi is} + e^{-4\pi is} K_-$, and move the integral of the entire function F to the path Π_0 that runs from x_c vertically to $\operatorname{Im} x = \frac{3\pi}{8}$ and then horizontally to $+\infty$ through x_0 . Summing over the two branches,

$$1 - \Sigma_1 = J_0 + J_- + R_- + R_+, \quad J_0 = \int_{\Pi_0} F ds, \quad J_- = \int_{\Gamma_-} F e^{-2\pi is} ds, \\ R_- = \int_{\Gamma_-} F e^{-4\pi is} K_- ds, \quad R_+ = \int_{\Gamma_+} F K_+ ds.$$

This is exact; $R_{\mp}$ carry all the terms $e^{2\pi ins}$ with $n \leq -2$ and $n \geq 1$ through the kernel. All three paths cross $\operatorname{Re} x = 0$ on the lines $\operatorname{Im} x \equiv \frac{\pi}{8} \pmod{\frac{\pi}{4}}$, where Lemma 35(ii) is sharp. We checked the contour identity numerically at $r = 0.2$ and 0.1 to 14 digits (`ident.py`).

Step 2: the landscape. A factor $e^{2\pi ins}$ has modulus $e^{-n\pi \operatorname{Im}(x e^{-i\theta})/r}$ on either branch, so the leading exponent of $|F e^{2\pi ins}|$ is $W_n(x) = \operatorname{Re}((\Phi_i(x) + i\pi n x) e^{-i\theta})$ for $\operatorname{Re} x \geq 0$, and at most $\Upsilon(x) - n\pi \operatorname{Im}(x e^{-i\theta})$ for $\operatorname{Re} x \leq 0$. Let $T_{\text{LO}} = -0.072274$ be the minimum of T_i over the θ -range. The following are certified in interval arithmetic by `certify_landscape_i.py` (mpmath iv, 30 s), for all θ in the range and all admissible x_c : the leading exponent is at most $T_{\text{LO}} - 0.1002$ on the start box, on Π_0 outside $|x - x_0| \leq 0.4$ ($n = 0$), on Γ_- outside $|x - x_-| \leq 0.4$ ($n = -1$), on Γ_- for $n = -2$ and on Γ_+ for $n = 1$, including the tails, by (iii); on the two windows $\operatorname{Re}(\Phi''_i e^{-i\theta}) \leq -0.682$; the kernels are at most 2.539 by (iv); and on all paths $d \geq 0.244$, $d' \geq 0.254$. With Lemma 35 each competing piece is at most $e^{(T_{\text{LO}} - 0.05)/r}$ for $r \leq r_0 = 1.56 \cdot 10^{-3}$.

Step 3: Laplace's method at the two saddles. On the window $x_0 + [-0.4, 0.4]$ the certificate gives $d \geq 0.4629$, so by Lemma 35(i) $|E| \leq 41.9|t|$ there, for both branches. Since $ds = dx/(2t)$ on both branches,

$$\int_{\text{window}} F ds = \frac{1}{2t} \left(\frac{2t}{\pi}\right)^{1/2} e^{i\pi/8} \int_{\text{window}} e^{\Phi_i(x)/t} (e^{h_+(x)} + i e^{h_-(x)}) dx + \mathcal{E},$$

summed over the branches, where $|\mathcal{E}|$ is at most $(e^{41.9r} - 1)$ times the same expression with absolute values inside. On the window $\text{Re}(\Phi_i(x)e^{-i\theta}) \leq \text{Re}(V_0e^{-i\theta}) - 0.341(x - x_0)^2$, so $\mathcal{E}$ is $O(|t|)$ relative to $e^{\text{Re}(V_0e^{-i\theta})/r}$. The remaining integral has an analytic integrand and a non-degenerate saddle at x_0 with $\Phi_i''(x_0) = -\frac{\sqrt{15}}{2}$, and the path is horizontal there. Laplace's method with a complex large parameter, uniform for $\arg t$ in a compact subset of $(-\frac{\pi}{2}, \frac{\pi}{2})$ [23], gives $(2\pi t \cdot \frac{2}{\sqrt{15}})^{1/2} e^{V_0/t} (e^{h_+(x_0)} + ie^{h_-(x_0)})(1 + O(t))$. The endpoints of the window contribute $O(e^{(\text{Re}(V_0e^{-i\theta}) - 0.054)/r})$. The powers of t cancel, and $J_0 = A_0 e^{V_0/t} (1 + O(t)) + O(e^{(T_{\text{Lo}} - 0.05)/r})$ with $A_0 = (2/\sqrt{15})^{1/2} e^{i\pi/8} (e^{h_+(x_0)} + ie^{h_-(x_0)})$. For J_- the factor $e^{-2\pi i s}$ is $e^{-i\pi x/t}$ on the even branch and $-e^{-i\pi x/t}$ on the odd branch. The saddle is x_- , and the same argument gives $A_- = (2/\sqrt{15})^{1/2} e^{i\pi/8} (e^{h_+(x_-)} - ie^{h_-(x_-)})$. With Step 2 this is (21).

The amplitudes. Put $z = e^{-2x}$. Then $h_- - h_+ = \frac{1}{2}(\log(1+iz) - \log(1-iz) + \log(1-z) - \log(1+z))$. At x_0 and at x_- , $|z| = a_0\sqrt{2} < 0.36$, so each logarithm has imaginary part below 0.37 in modulus, and $e^{h_- - h_+}$ is the square root with positive real part of $\frac{(1+iz)(1-z)}{(1-iz)(1+z)}$. At $z = -a_0(1+i)$ this ratio is $\frac{1+2a_0+2a_0^2}{1-2a_0+2a_0^2}$. With $2a_0 = \sqrt{5} - \sqrt{3}$ and $2a_0^2 = 4 - \sqrt{15}$ one checks $\phi(1 - 2a_0 + 2a_0^2) = 4a_0 = \phi^{-1}(1 + 2a_0 + 2a_0^2)$, so the ratio is ϕ^2 and $e^{h_- - h_+} = \phi$ at x_0 . At x_- , $z = a_0(1+i)$, the ratio is ϕ^{-2} and $e^{h_- - h_+} = \phi^{-1}$. This gives the stated A_0 and A_- , and $1 + i\phi \neq 0$, $1 - i/\phi \neq 0$. For the ratio, $h_+(x_-) - h_+(x_0) = \frac{(1+i)\pi}{4} + h_-(x_0) - h_+(x_0)$, because $e^{-2x_-} = -e^{-2x_0}$ and the logarithmic part of h_+ at $-z$ is that of h_- at z . So $e^{h_+(x_-) - h_+(x_0)} = \phi e^{(1+i)\pi/4}$. Also $(1 - i/\phi)/(1 + i\phi) = -i/\phi$, since $\phi + \phi^{-1} = \sqrt{5}$ and $1 + \phi^2 = \frac{5+\sqrt{5}}{2}$. Hence $A_-/A_0 = -ie^{(1+i)\pi/4} = e^{(1-i)\pi/4}$.

Step 4: Rouché. Put $u = 1/t$ and $\varrho = A_-/A_0$. The zeros of $1 + \varrho e^{\Delta V u}$ are the solutions of $\Delta V u = -\frac{(1-i)\pi}{4} + i\pi(2m+1)$, $m \in \mathbb{Z}$. Those with $\text{Re } u > 0$ and $\arg u \rightarrow -\theta_i$ are $m = -j - 1$, $j \geq 0$, that is $u = u_j$. For $t \in \overline{D_j}$ we have $|t| \leq 1.15 \cdot 10^{-3}$ and $|\arg t - \theta_i| \leq 10^{-3}$ once $j \geq 285$. Here $\arg t_j^0 - \theta_i = \arctan \frac{2}{16j+6}$, the disc subtends an angle $O(1/j)$, and both extremes are attained on ∂D_j , where we evaluated them. So for $j \geq 285$ every $t \in \overline{D_j}$ satisfies the hypotheses of Steps 1–3. On D_j the function $A_0 e^{V_0/t}$ is analytic and non-zero, and

$$\frac{1 - \Sigma_1}{A_0 e^{V_0/t}} = g_0(u) + O(1/j), \quad g_0(u) = 1 + \varrho e^{\Delta V u} = 1 - e^{\Delta V(u - u_j)},$$

using $|\varrho e^{\Delta V u}| \leq e$ on D_j , $|e^{V_-/t}/e^{V_0/t}| = |e^{\Delta V u}|$ and $e^{(T_{\text{Lo}} - 0.05)/r}/|e^{V_0/t}| \leq e^{-0.05/r}$. As in Step 5 of the proof of Proposition 32, $|g_0| > 0.28$ on ∂D_j , where $|\Delta V(u - u_j)| = 1$, and u_j is the only zero of g_0 in D_j , a simple one, because the zeros of g_0 are $2\pi/|\Delta V|$ apart. By Rouché's theorem $1 - \Sigma_1$ has exactly one zero in D_j for large j , and it is simple. From $|g_0(u^*)| = O(1/j)$ we get $u^* - u_j = O(1/j)$, that is $t_j^* - t_j^0 = O(j^{-3})$. The map $t \mapsto ie^{-t}$ is conformal, so the zero is simple in q .

Step 5: the poles of G_0^T . Step 6 of the proof of Proposition 32 applies at every zero of $1 - \Sigma_1$ in $0 < |q| < 1$: $\Sigma_0 S_0 = 2q/(1 - q) \neq 0$, so $\Sigma_0(q_j^*) \neq 0$ and q_j^* is a simple pole of G_0^T . The coefficients of $1 - \Sigma_1$ and Σ_0 are real, so the same holds at $\bar{q}_j^*$. Finally $\sqrt{q_j^*} \rightarrow \pm e^{i\pi/4}$ and $\sqrt{\bar{q}_j^*} \rightarrow \pm e^{-i\pi/4}$. $\square$

The constants in Steps 2 and 4 are explicit. The $O(t)$ of Laplace's method is not made explicit, so j_0 is not either. Numerically (`rootunity_zeros/amp_i.py`), the ratio of $1 - \Sigma_1$ to the two-saddle sum in (21) minus 1 is $0.0143 + 0.0107i$, $0.0088 + 0.0008i$ and $0.0064 + 0.0043i$ at $r = 0.04, 0.02, 0.01$ on the ray $\theta = \theta_i$. The predicted zeros are accurate far below j_0 : at 80–430 digits (`zeros_i.py`), $|t_j^* - t_j^0| j^3 = 0.0092, 0.0060, 0.0065, 0.0065$ at $j = 1, 10, 40, 80$. The first zero is $q_1^* = 0.1404 + 0.8293i$.

Corollary 37 (the poles of U and V accumulate at $x = \pm e^{\pm i\pi/4}$). *For all sufficiently large j , U and V have poles at $x = \pm \sqrt{q_j^*}$ and at $x = \pm \sqrt{\bar{q}_j^*}$, with q_j^* as in Proposition 36. These tend to the*

four primitive eighth roots of unity $\pm e^{\pm i\pi/4}$, that is to $\pm\sqrt{i}$ and $\pm\sqrt{-i}$. At $q = q_j^*$ we have $S_e \neq 0$,

$$t_1 = t_1^* + O(1/j), \quad t_1^* = \frac{(\sqrt{15} - 3) - (\sqrt{15} + 3)i}{12} = 0.07275 - 0.57275i,$$

$$1 - \mathfrak{g}_U t_1 = \frac{(21 - \sqrt{15}) - (\sqrt{15} - 3)i}{24} + O(1/j), \quad 1 - \mathfrak{g}_V t_1 = \frac{3}{4} - \frac{\sqrt{15}}{12}i + O(1/j),$$

and Π_1, Π_q tend to non-zero limits at both roots: $\Pi_1 \rightarrow -3.0048 + 2.9237i$, $\Pi_q \rightarrow -2.7091 + 3.3891i$ at the root tending to $e^{i\pi/4}$, and $\Pi_1 \rightarrow 0.2866 + 0.2945i$, $\Pi_q \rightarrow -0.2287 - 0.7363i$ at the root tending to $-e^{i\pi/4}$. (Standing: Theorems 12 and 15.)

Proof. Step 1: $S_{\pm}$ near $q = i$. On the two branches, $b_{2s}q^{\pm 2s} = (-1)^s e^{\mp x} f_e(s)$ and $b_{2s+1}q^{\pm(2s+1)} = \pm i(-1)^s e^{\mp x} f_o(s)$, with x the branch coordinate. So on each branch $S_{\pm}$ is $\sum_s (-1)^s G(s)$ with G entire and vanishing at the negative integers. Use the kernel $1/(2i \sin \pi s)$, which equals $e^{-i\pi s}/(1 - e^{-2\pi i s}) = e^{-i\pi s} + e^{-3\pi i s}/(1 - e^{-2\pi i s})$, with the same s_c and paths $\Gamma_{\pm}$ as in Step 1 of the proof of Proposition 36. Move the integral of the entire term $Ge^{-i\pi s}$ to the path Π_S that runs from x_c vertically to $\text{Im } x = \frac{\pi}{8}$ and then horizontally to $+\infty$. This gives the exact split $S_{\pm} = J_S + R_S^- + R_S^+$, with $J_S = \int_{\Pi_S} Ge^{-i\pi s} ds$, $R_S^- = \int_{\Gamma_-} Ge^{-3\pi i s}(1 - e^{-2\pi i s})^{-1} ds$ and $R_S^+ = \int_{\Gamma_+} Ge^{i\pi s}(1 - e^{2\pi i s})^{-1} ds$. (The contour identity was checked numerically at $r = 0.2$, `ident_S_i.py`.) The leading exponents are now W_{ν} with $\nu = -\frac{1}{2}$ on Π_S , $\nu = -\frac{3}{2}$ on Γ_- and $\nu = \frac{1}{2}$ on Γ_+ . The main saddle is $y_0 = x_0 - \frac{i\pi}{4}$: there $e^{-4y_0} = -i(4 - \sqrt{15})$, so $\Phi'_i(y_0) = \frac{i\pi}{2}$ exactly, and $\Phi''_i(y_0) = -\frac{\sqrt{15}}{2}$. Let $E_S = \Phi_i(y_0) - \frac{i\pi y_0}{2}$ and $T_S = \text{Re}(E_S e^{-i\theta}) \in [0.42106, 0.42198]$. The certificate `certify_landscape_S_i.py` (same machinery, 21 s) proves that the leading exponent is at most $T_S - 0.1065$ on every competing piece (start box, Π_S outside $|x - y_0| \leq 0.4$, $\Gamma_{\pm}$, tails), that $\text{Re}(\Phi''_i e^{-i\theta}) \leq -0.682$ on the window, and that $d \geq 0.4629$ there. The factor $e^{\mp x}$ is $O(1)$ in the exponent. On the odd branch the kernel term is $e^{-i\pi s} = ie^{-i\pi x/(2t)}$, so the odd weight is $(\pm i) \cdot i \cdot ie^{h-} = \mp ie^{h-}$. Laplace's method as in Step 3 gives, for $r \leq 1.557 \cdot 10^{-3}$,

$$S_{\pm} = \left(\frac{2}{\sqrt{15}}\right)^{1/2} e^{i\pi/8} e^{\mp y_0} e^{h_+(y_0)} (1 \mp e^{2\pi i/3}) e^{E_S/t} (1 + O(t)).$$

Here $ie^{h_-(y_0) - h_+(y_0)} = e^{2\pi i/3}$. Indeed $z = e^{-2y_0} = a_0(1 - i)$, the ratio $\frac{(1+iz)(1-z)}{(1-iz)(1+z)}$ equals $w/\bar{w}$ with $w = (\sqrt{15} - 3) + (\sqrt{5} - \sqrt{3})i$, and $\arg w = \frac{\pi}{6}$ because $\sqrt{3}(\sqrt{5} - \sqrt{3}) = \sqrt{15} - 3$; the root with positive real part is $e^{i\pi/6}$. Since $1 - e^{2\pi i/3} \neq 0$, $S_e = S_+ \neq 0$.

Step 2: t_1 . Step 0 of the proof of Proposition 33 holds on $|q| < 1$, so $t_1 = -\frac{1}{2}(1 + S_-/S_e)$. By Step 1, $S_-/S_e = e^{2y_0} \frac{1+e^{2\pi i/3}}{1-e^{2\pi i/3}}(1 + O(t)) = e^{2y_0} \frac{i}{\sqrt{3}}(1 + O(t))$, and $e^{2y_0} = \frac{\sqrt{5}+\sqrt{3}}{\sqrt{2}} e^{i\pi/4}$. So $S_-/S_e = \frac{3+\sqrt{15}}{6}(-1+i) + O(t)$ and $t_1 = t_1^* + O(t)$. At q_j^* , $t = t_j^*$ satisfies the hypotheses for large j (Step 4 of the proof of Proposition 36), and $|t_j^*| \asymp 1/j$.

Step 3: the factors. As $q \rightarrow i$, $\mathfrak{g}_U = \frac{q}{1-q^2} \rightarrow \frac{i}{2}$ and $\mathfrak{g}_V = \frac{q}{1-q} \rightarrow \frac{-1+i}{2}$. This gives the two displayed limits, which are non-zero; the second has modulus $\sqrt{2/3}$. So $B_0 = 1/(1 - \mathfrak{g}_U t_1)$ is finite and non-zero at q_j^* for large j . The first two paragraphs of the proof of Corollary 34 apply at q_j^* without change: they need only $0 < |q_j^*| < 1$, $S_e(q_j^*) \neq 0$ and $1 - \mathfrak{g}_U t_1 \neq 0$. So $x^* = \pm\sqrt{q_j^*}$ is a pole of $\mathcal{W}(\cdot, y)$ as soon as $\Pi_y(x^*) \neq 0$, and (10)–(11) hold at q_j^* for both roots. In the limit $q = i$, $x = \pm e^{i\pi/4}$, every factor of these closed forms is an explicit algebraic number: $\frac{2q(1+x)}{1-q}$ and $\frac{1+x}{1-x}$ are finite and non-zero because $x \neq \pm 1$; the real part of $t_1^* + \frac{1+x}{2x}$ is $\frac{\sqrt{15}-3}{12} + \frac{1}{2} + \frac{\sqrt{2}}{4} > 0$ at $x = e^{i\pi/4}$, and its imaginary part is $\frac{3\sqrt{2}-3-\sqrt{15}}{12} \neq 0$ at $x = -e^{i\pi/4}$; and $t_1^* \frac{1-x+q}{1+q} + \frac{1}{2x}$ has modulus above 0.64

at both roots. The last bound and the four limit values above are enclosed in interval arithmetic by `rootunity_zeros/factors_i.py`. So $\Pi_1, \Pi_q \neq 0$ at both roots for large j .

Conclusion. For large j , $\pm\sqrt{q_j^*}$ are poles of V and of U . The coefficients are real, so the conjugate points are poles as well, and $\sqrt{q_j^*} \rightarrow \pm e^{i\pi/4}$. $\square$

Numerically (`zeros_i.py`, `tlim_i.py`): S_+ divided by the right side of Step 1, minus 1, is $0.0261 + 0.0022i$ and $0.0128 + 0.0010i$ at $r = 0.02, 0.01$. At the zeros, $t_1(q_j^*) = 0.0599 - 0.5540i$, $0.0663 - 0.5632i$ and $0.0696 - 0.5680i$ at $j = 20, 40, 80$, and $(t_1 - t_1^*)/t_j^* \approx 0.08 + 1.42i$. At $q_1^* = 0.1404 + 0.8293i$ and $q_2^* = 0.0802 + 0.8974i$ the closed forms (10)–(11) agree with the pairings computed from truncated operators ($N = 195$ and 311) to relative error at most $1.3 \cdot 10^{-33}$, for $y \in \{1, q\}$ and both roots (`pairing_check_i.py`).

Remark 38 (where on the circle the poles accumulate). Theorem 31 does not locate the singularities on the circle. The real travel poles accumulate only at $x = \pm 1$ (at $q = 1$ for G_0^T). By Proposition 32, the poles of G_0^T also accumulate at $q = -1$, along the two rays $\arg t = \pm \arctan(\pi/\log 4)$ in $q = -e^{-t}$, that is at $x = \pm i$. This is proved unconditionally, with the computer-assisted part confined to the interval certificate of Step 4. Every point of the circle is singular. Numerically the pole set is much larger. The function $1 - \Sigma_1 = \cos(Z; q^2)$ has 64 zeros in $|q| < 0.94$ by the argument principle: the travel poles q_1, q_2 and 62 non-real zeros in 31 conjugate pairs, the first at $q = -0.4075 \pm 0.5449i$. By Step 6 of the proof of Proposition 32, every zero of $1 - \Sigma_1$ in $0 < |q| < 1$ is a pole of G_0^T . For U one also needs $\Pi_q \neq 0$ there, together with $S_e \neq 0$ and $1 - \mathfrak{g}_U t_1 \neq 0$. At each of the 62 non-real zeros Π_q was found non-zero at 40 digits; this is numerical. At the zeros q_j^* of Proposition 32 all three are proved for large j , by Proposition 33 and Corollary 34. So the poles of U and of V accumulate at $x = \pm i$, along the images of the rays $\arg t = \pm \theta_1$, with the standing of Theorems 12 and 15. The computer-assisted inputs are the two landscape certificates. The same holds at $q = \pm i$. By Proposition 36 the poles of G_0^T accumulate at $q = i$ along the ray $\arg t = \theta_i = \arctan(\pi/(6 \log 2))$ in $q = ie^{-t}$, and at $q = -i$ along the conjugate ray. By Corollary 37 the poles of U and V accumulate at the four primitive eighth roots of unity $x = \pm e^{\pm i\pi/4}$, with the same standing. There two saddles of equal height compete, and the computer-assisted inputs are the certificates `certify_landscape_i.py` and `certify_landscape_S_i.py` and the interval evaluation `factors_i.py`. So the poles of U and V are proved to accumulate at $x = \pm 1, \pm i$ and $\pm e^{\pm i\pi/4}$. Plotted, the zeros approach roots of unity along straight lines. We conjecture that the poles are dense on the circle. A route is the same resummation at $q = \zeta e^{-t}$ for other roots of unity ζ ; Propositions 32 and 36 are the cases $\zeta = -1$ and $\zeta = \pm i$. For other ζ nothing is claimed. The natural boundary does not depend on any of this.

7 Open questions

The travel resolvent $\Sigma_0/(1 - \Sigma_1)$ is transcendental, not D -finite, and has $|q| = 1$ as a natural boundary, unconditionally. The growth series U of W_{univ} and its relaxed companion V have the same properties in x , on the transfer model (Theorem 12) and the junction pairing (Theorem 15); two finite ranges are computer-assisted, the six gate poles above $\tau = 5 \cdot 10^{-3}$ and the positivity of the pairing at the first twelve poles, and neither is needed for transcendence, only for the statements “at every travel pole”. The series in question is the growth series of the universal group W_{univ} ; for a particular right triangle it agrees with that of W_T only to the effective universality depth of [3]. Several questions remain open.

- (1) Are the poles of U dense on the unit circle (Remark 38)?

- (2) Are U and V D -algebraic? The pole argument does not decide this.
- (3) Is the number β_2 transcendental? The question reduces to a value statement for a Hahn–Exton q -Bessel function, and the obstruction sits at the arithmetic ratio $\rho = \frac{1}{2}$ [5, Appendix app:beta2].
- (4) Is $(u_n \bmod p)$ non-automatic for some prime p ? By Christol’s theorem this would give a second, analysis-free proof of transcendence; the p -kernel is computed to be maximally non-automatic at $p = 3, 5$ [5, §sec:modp].
- (5) The number of travel poles in a window of w , equivalently an asymptotic count of the zeros of $\cos(\cdot; q^2)$ in $(0, Z(q))$ uniform as $q \rightarrow 1$, is not known [5, Remark “what the known theory does not give”, rem:knowntheory].

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